claude-windows / 9f6357d8-0e35-491a-a830-052e094c7f37

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\subagents\workflows\wf_b6ef6275-de5\agent-a711967821136d6a9.jsonl`
- SHA-256: `39feaa8c830ec6d7f6ea159eeb944832ebb7c72050db0a08c457447ddc3ac3b3`
- Source modified: `2026-06-21T13:58:45+00:00`
- Imported at: `2026-07-05T16:48:25+00:00`
- project: `wf_b6ef6275-de5`
- session_id: `9f6357d8-0e35-491a-a830-052e094c7f37`

Transcript

1. user (2026-06-21T13:56:13.270Z)

Read the RALLY-DETECTION QUALITY track docs in the engine repo (C:/Users/avidu/Projects/badminton-highlight-indexer) and extract status + next steps. Read docs/RALLY_QUALITY_RESEARCH.md (esp. its §7 progress tracker), docs/HOW_RALLY_DETECTION_WORKS.md, docs/MULTI_SIGNAL_FUSION_PLAN.md (the held-shuttle Gate-A rally_gate.py not wired into production = cheapest win), docs/EXPERIMENT_HARNESS.md, docs/GOLDEN_REGRESSION_FIXTURES.md, docs/ROORA_LABELING_GUIDELINES.md, and docs/NEXT_STEPS.md (the prioritized pick-up for this track). Capture the prioritized next steps (R2 metric ? R4 end-rule, Gate-A wiring, golden-set growth, etc.), gates, effort. Return structured JSON (track="rally-quality").

2. assistant (2026-06-21T13:56:16.232Z)

I'll start by pulling the repo to ensure freshness, then read the rally-quality track docs.

3. user (2026-06-21T13:56:19.661Z)

fatal: Cannot fast-forward to multiple branches.

4. user (2026-06-21T13:56:19.880Z)

```
ABLATION_LAYER.md
ANALYZERS_AND_RECONCILERS.md
BACKLOG.md
CLOUD_GPU_DRYRUN_GUIDE.md
CODE_AUDIT_AND_TEST_HARDENING.md
CODE_MAP.md
COMMERCIALIZATION.md
COMPUTE_DECOUPLED_SERVING
DEFERRED_HARDENING_PLAN.md
DESKTOP_APP_PLAN.md
DOC_STATUS.md
EVALUATION.md
EXECUTION_POLICY.md
EXPERIMENT_HARNESS.md
GOLDEN_REGRESSION_FIXTURES.md
HARDENING_LOOP_HANDOFF.md
HOSTING_PLAN.md
HOW_RALLY_DETECTION_WORKS.md
I18N_PLAN.md
KHELSUTRA_PRODUCT_OVERVIEW.md
MULTI_SIGNAL_FUSION_PLAN.md
NEXT_STEPS.md
OWNED_MODEL_IMPLEMENTATION_PLAN.md
PERSONALIZATION_PLAN.md
```

PER_VIDEO_WORKSPACE.md
PLATFORM_ARCHITECTURE.md
POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md
PROJECT_DOC_TEMPLATE.md
PROMINENT_COURT_DETECTION.md
QUALITY_ITERATIONS.md
R1_MILESTONE_LAUNCH_REPORT.md
R2_EVAL_METRIC_DESIGN.md
RALLY_DETECTION_GAP_CLOSURE.md
RALLY_DETECTION_QUALITY_REPORT.md
RALLY_QUALITY_RESEARCH.md
README.md
REAL_FOOTAGE_VALIDATION_REPORT.md
ROADMAP.md
SETUP_NEW_MACHINE.md
STORAGE_SHARING_MODEL.md
TRACKNET_WSL_SETUP.md
UI_ALPHA_EXECUTION_PLAN.md
UI_PROPOSAL.md
VIDEO_LOCALITY_MODEL.md
archives
data_pipeline
mockups

5. assistant (2026-06-21T13:56:22.476Z)

The branch has multiple upstreams; I'll proceed read-only since I cannot cleanly fast-forward. Let me read the rally-quality track docs.

6. user (2026-06-21T13:56:23.473Z)

1 # Next steps (live) ? refreshed 2026-06-21
2
3 > **How to read:** this top section is the **current cross-track state + prioritized pending list** (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed **owned-model & rally-quality track** plan (2026-06-13) is **preserved below, still valid** ? it is now *one of several* active tracks. Live blow-by-blow: memory `current-state.md`. Doc lifecycle/hygiene: `docs/DOC_STATUS.md`.
4
5 ## ? Where we are (2026-06-21)
6 **DECOUPLED_COMPUTE delivered on GCP + archived** (#270); **compute = GCP only** (ADR-013, DO dropped). **`COMPUTE_DECOUPLED_SERVING`** subproject spun up (#272, **ADR-014**) ? one pure core + local/cloud profiles; **owner M0 design sign-off pending**. **Per-video workspace P0** shipped (#264); **doc-status register + archival due-diligence** in place (#273). The **khelsutra site + per-rally UI** ship in parallel (sibling track). The **rally-quality / owned-model** track (below) is the core ML work, gated on **growing the golden corpus**.
7
8 ## ? Prioritized pending ? all tracks
9 **P0 ? highest leverage / unblocks the most**
10 1. ? **`COMPUTE_DECOUPLED_SERVING` M0 approved + M1?M4 SHIPPED (#279/#280/#282/#283) + M5+ batch authorized** (owner 2026-06-21, ADR-014 ratified). All §8 gating Qs locked (D7?D17). **Owner residual** (not blocking Claude's default-OFF code): counsel ToS/DPA (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF team-domain/AUD + ingress-lock; real cloud verification authorized within a **$100/?10k cap** (D17).
11 2. **[owner, ongoing] Grow the golden corpus** ? multi-court badminton + ?3 matches/sport (TT first). The single highest-leverage move; the quality track *and* full-corpus training feed on it.
12 3. **[owner + Claude · GPU] Full-corpus training (n?5)** ? a real `weights/<ver>/` (the `0.1.0-l4` bundle is n=2 plumbing-only).
13
14 **P1 ? near-term, mostly Claude-doable (default-OFF, green-gated)**
15 4. **[Claude] `COMPUTE_DECOUPLED_SERVING` M5?M12 default-OFF code batch** (owner-authorized 2026-06-21, merge-each-on-green): M5 truly-local preset+startup-checks+L4 parity test ? M6 `backend/compute/` Cloud Run dispatch + Dockerfile (mocked) ? M8 v6 ledger+pricebook+/cost ? M9-auth Cf-Access JWT+tenant-scoping ? M11 flywheel plumbing ? M12 `gdrive-api` backend.

(M1?M4 config/input/output/device layer = ? SHIPPED.)

5. **[Claude] Per-video workspace P1?P6** (P1 v6 DB migration ? P2 worker ? P3 per-video routing of `detector/`+`tmp/` *[the `output/chunks_*` de-hardcode base = ? done via #282]* ? P4 ? P5 Launch-Report flip ? P6 GPU parity). ? P1's v6 migration coordinates with the M8 v6 cost-ledger migration ? author the version bump once.
6. **[Claude] Ops-Agent codification into `deploy/gcp`** (every VM auto-enables observability ? [[gcp-ops-agent-enable]]).
7. **[owner] Set the ship-gate target (#172)** ? needs corpus growth.
8. **[owner-side GPU] R-series: promote-or-reject `R2` tolF1 as the gate** (the ablation) ? needs GPU-extracted golden features.

P2 ? later / gated / deferred

9. Rally-quality **P3/P4** (person-fusion LOVO, audio cue) ? needs GPU golden features.
10. **M0.4 scorer swap** ? TemporalMaxer (per the 2026-06-17 external review).
11. **Multisport** (table-tennis first, then pickleball).
12. Serving **M8/M9** ? per-video cost ledger, edge auth, DPA/consent (within `COMPUTE_DECOUPLED_SERVING`; owner/legal-gated).
13. **WASB-C3** weight provenance (gates *sharing* a trained model) ? legal.
14. **Rename repo off "badminton"** (owner, BACKLOG).
15. Owner housekeeping: retire the shared **DO droplet** + **DO token**; collector **`md5`** export (collector repo).
16. **PMF field validation** (owner-side, business track).

Doc-hygiene (tracked in `docs/DOC_STATUS.md` §4): ? #1 per-video reconcile + ? #2 this refresh; **pending:** field/PMF/Roora ? archives + a data-layer `docs/` dir (owner boundary confirm); confirm the COMPLETED top-level stubs; verify `HOSTING_PLAN`/`UI_PROPOSAL` currency.

Owned-model & rally-quality track ? detail (2026-06-13 plan; still valid)

Updated: 2026-06-13 (**rally-detection quality track**: fusion + experiment-harness design docs merged, owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior: 2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**: ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-first delivery ? see `docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**; **PR-2 upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload MD5 byte-identity + compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep #90?#98 landed** ? ruff/mypy/ coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin collapse into `trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626 re-verified exact).
Authoritative roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-blow:** memory `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do next," refreshed each session.

? Next-session / GPU-box pickup (2026-06-15)
M0 in-container quick wins are COMPLETE: ship-gate provisional blocker (**#170**), train/eval split guard (**#171**), the **doc-consistency sweep** (**#174** ? corrected 16 drifted docs vs the code), and the **P3 person-feature fusion litmus machinery is already present + suite-green** (`backend/eval/fusion_compare.py` + `ablation.py`: person-driven ?F1 / bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted data). ? CI is red repo-wide due to a **GitHub Actions minutes/billing outage** (account-level, not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing ? Actions minutes / spending limit).

Immediate next step ? the real P3 LOVO measurement (this box has a GPU + `torch+cu124` + `cv2`; videos and trajectory CSVs are present, but the GPU-extracted golden

features + `.rallies.csv` labels are not yet here):
1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-data\features\2026-06-15-n6"` ? produces `*_golden_features.json` for the 6 golden videos. (GPU + WSL + videos + trajectory CSVs + `.rallies.csv` labels are owner-side only.)
2. Artifacts land in the **shared cloud folder** `C:\Users\avidu\OneDrive\rally-annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow: **[`GOLDEN_DATA_SHARING.md`](data_pipeline/GOLDEN_DATA_SHARING.md)**; tracked in **#175**.
3. **[CPU dev/agent session]** `python -m backend.eval.fusion_compare --features-dir <shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir <shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to also pass the `skipif` litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`).

Also queued (owner-gated / pending input):
- Ship-gate target calibration ? **#172** (owner decision; needs corpus growth).
- Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent will do it).
- Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config bug.

Leaving (bulky / M2 ? no start without owner go): ActionFormer (M0.4), event-index DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack deprecation, multisport, repo rename, PMF execution.

Where we are (one paragraph)
ADR-008 ratified ("repo split driven by productionization, not isolation"): training lives in-repo (`training/`)
behind a CI-enforced import wall; the featurizer is single-sourced (`backend/features/rally_seq.py`, train==serve);
the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest output/human_lovo_manifest.json --floor
eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0 (LOVO **0.626**, floor passed,
sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The **n=6 owned corpus is in the collector**
(262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet reclassification). The **Gemini
battery is done** (see table) ? the moat is quantified.

The quality table that now drives strategy (IoU 0.5 vs human golden labels)
Path	Mahadevpura (clean-ish)	Kushagra (multi-court)
Heuristic (velocity windowing)	0.657	0.157
Gen-0 owned head (LOVO, n=6)	0.744	0.561
Two-stage WASB+Gemini refine	0.83	?
Gemini wrapper (`gemini-flash-latest`)	**0.93**	**0.66**

Reading: clean footage is commoditized (serve it with Gemini for cents). **Multi-court drops the commodity
wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game pickups + dead-time merges,
the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-growth-sized gap, not an
architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG tapes (uploads candidate clips
only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry load-bearing.
Baselines tracked in `eval_baselines/` (heuristic_lovo_n6.json, gemini_wrapper_2026-06-10.json).

Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation in progress)
This is the track a desktop/local agent is slated to pick up. Design docs landed (PR #112 + session). All implementation is **owner-gated** (per explicit in the plan); each step = ONE PR off `master`.

90

91 **Hardening loop (T1-T5 positive tests + audit + finale) COMPLETE (2026-06-14, PRs #156/#157/#160/#161/#163 + #165 + final records).** See archived `docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` (Living Log with every pre/post/review) and `docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` (full rules + STATE). Top-level now pointer stubs per #165 review. All 5 items + verification + archive done; clean slate.

92

93 - **Docs:** `docs/MULTI_SIGNAL_FUSION_PLAN.md` (shuttle + person + audio, one fusion layer) ·

94 `docs/EXPERIMENT_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) ·

95 `docs/ROORA_LABELING_GUIDELINES.md` (v8) · `docs/HOW_RALLY_DETECTION_WORKS.md` (2-layer mental model).

96 - **Key finding (no re-investigation needed):** the owner's "held-shuttle counted as a rally" bug is **NOT a missing capability** ? `backend/pipeline/detectors/rally_gate.py` already implements the net-crossing **"Gate A"** (`apply_gate(..., min_crossings=2)`) that kills service-feed/held-shuttle windows, but it is **only called from eval drivers** (and there was no `min_crossings` knob in `IndexingConfig`). **Wiring Gate A into the live segmenter (now via fusion_hybrid) was the single cheapest, highest-confidence quality win.** (P2 FusionSegmenter specifics: registry `fusion_hybrid`, **default-OFF**, seeds from substrate, persists `net_crossings` + optional person features, optional served Gate A/B via config knobs; adversarially reviewed; suite green. See `tests/test_served_gate_a_regression.py` for the honest finding that on production windowing Gate A is neutral/negative in some cases, hence kept opt-in with knob for safety, and the leaderboard revert note.)

97 - **Current status (2026-06-14):** P1 (person cue extraction to reusable `PersonDetector`) + P2 (FusionSegmenter + Gate A signal + served handoff gating) + experiment harness P1 DONE (adversarial review + NO-REGRESSION; suite green). **P3 next (this PR + follow-ups):** learned fusion (person features ? harness `compare` LOVO lift on the 6 golden videos; eager-person-compute optimisation); then P4 audio via hit_detector.

98 - **Pick-up order (each = ONE PR off `master`, owner approval first, flag-gated default-OFF, promote only on measured LOVO):**

99 1. (P2 first-step, largely done via fusion_hybrid) Wire Gate A into production served path + config knob (min_crossings); apply in hybrids at runtime. (Now available opt-in; served neutral on production windowing per litmus ? kept opt-in with knob for safety.)

100 2. P3 first step (this work item): harness `compare` LOVO litmus for person-on variant (players_in_court + two_sided + colocation etc. from `fusion_features`) on the 6 golden; add the eager compute optimisation note/landing. Pure + harness (testable here).

101 3. P3/P4 continuation: audio cue integration + full learned promotion if lift proven.

102 - **Discipline (from EXPERIMENT_HARNESS.md):** every new cue **flag-gated, default OFF**; promote to default **only on measured LOVO lift** through `run_regression.py` (exit 0/1/3); pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.

103 - ? Gate A, person cue logic, harness, and pure fusion paths are testable here; the heavier real-video / golden LOVO / ablation confirmation waits on the GPU-extracted golden features + `.rallies.csv` labels, which are not yet present in this checkout.

104 - **External due-diligence review inputs (2026-06-17) ? PICK UP AFTER current ideas are exhausted; full details + citations in [`RALLY_QUALITY_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md) §5.6.** An owner-commissioned 31-reference review of the quality report *validated* the methodology and surfaced these **future** levers (research inputs, not active work, not new claims; verify citations before any submission):

105 1. **"First-mile" paper framing (highest-value):** stroke-classifiers (BST/DSTA) + the big datasets (ShuttleSet ~36k strokes, Fine-Badminton ~27k actions) all **assume a pre-trimmed rally clip already exists** ? we solve the ignored prerequisite. Pitch the paper as *"the missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage."*

106 2. **Ground R2 in Action-Spotting / Precise-Event-Spotting** (SoccerNet/TenniSet/T-DEED) ? turns R2 into a defensible publishable contribution.

107 3. **Add domain baselines for any submission:** MonoTrack + BST on ShuttleSet / Fine-Badminton (we have only heuristic + Gemini today).

108 4. **Prefer TemporalMaxer over ActionFormer** for the M0.4 scorer swap (parameter-free max-pool, ~2.8× fewer GMACs, small-N robust) ? see "M0.4 next increment" below.

109 5. **Court polygons ? pose ? the rally-END lever:** masking on-court players can resolve *occluded end-of-rally states* (the #1 remaining gap vs Gemini), not just spectator noise.

110 6. **Name the eval?serve lesson "pipeline distribution skew"** ? a citable paper "lesson."

111

Discipline reminder: owner approval before starting any owner-gated item; one PR per step; babysit the PR (poll/fix/reply) until merge observed, *then* advance.

Prioritized next steps

1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move; every lever below feeds on it.
   Priority order for new videos: **(a) multi-court badminton** (the target distribution AND where the ship target lives), **(b) ?3 matches per new sport ? table tennis first** (WASB transfers at F1 0.909; pickleball labels are corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c) capture variety per `VIDEO_RECORDING_GUIDELINES.md`; **adult sessions only for the training pool** (consent guardrail). Per video: label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ? re-run the Gen-0 harness.
   The paired sign test needs n: at n=10 with 9 wins, p?0.011 ? significance is reachable this month.
2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480 (necessary), **multi-court reference = wrapper 0.66** (the number that makes the owned model worth serving), clean-footage ceiling = 0.93 (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70 on multi-court folds with paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.
3. **M0.4 next increment ($0/local):** swap the Gen-0 LogReg for a lightweight temporal head (MIT, minutes on the 2070S ? NOT the heavyweight Gen-2 stack) inside the existing harness; per-video threshold calibration (the LOVO-vs-oracle gap is visible per fold in the harness output). **Prefer TemporalMaxer** (parameter-free max-pool, ~2.8× fewer GMACs, small-N robust) over ActionFormer/ASFormer per the 2026-06-17 external review (`RALLY_QUALITY_RESEARCH.md` §5.6) ? better fit for the cheap/local/fast mandate.
4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`); merged-prototype LOVO across badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-on-WASB serving stays gated by C3 (provenance multiplies per sport).
5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+ the two added questions: "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot at the warmest 2-3 academies (consent at capture; demo served via the Gemini path or human-polish ? the reel pattern exists: `output/MahadevpuraSingles_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner ratified the physical-visit approach; reframed product-feedback-first per owner spec):** `docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md` (the recruiting flyer ? owner fills ?/contact placeholders and recruits) + `docs/FIELD_VISIT_PLAYBOOK.md` (the hired associate's visit playbook: coach + student script, demo-after-Q6 rule, ground rules, Phase-2 record-and-show-back stretch goal) + `docs/FIELD_VISIT_ANSWER_SHEET.md` (per-visit capture form incl. demo-reaction + student blocks) + `docs/PMF_FIELD_SIGNALS.md` (G/A/R anchors + **pre-registered decision rules** ? PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first visit).
6. **Housekeeping:** ~~rotate the Gemini API key~~ **DONE 2026-06-10** (owner-verified working; confirm old-key deletion in AI Studio); **rename repo off "badminton"** (owner action, in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the battery completed.

Pointers / artifacts

```
- Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Topology: ADR-008 (`docs/archives/decisions/DECISIONS.md`) ·
  Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` · Quality history:
  `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
- Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact bundle; weights gitignored,
  manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
- Corpus: collector DB (system-of-record, 6 matches/262 rallies) · eval manifest `output/human_lovo_manifest.json`
  (gitignored) · durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation Setup\Collect\Trajectories\` ·
  labels: `...\Collect\Golden Labelled\`.
- Gemini ops learnings: serial uploads (`indexing.ai_max_workers=1`) + chunk re-encode
  (`indexing.ai_chunk_scale_width=1280`) + 6-attempt exponential generate retry ? a cold/prepaid project burst-throttles
  with a MISLEADING "credits depleted" 429; ramp gradually.
- ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next milestone. Committed next
  PLATFORM slice = async job model (single-tenant).

```

7. user (2026-06-21T13:56:24.455Z)

```
# Rally-Detection Quality: Research, Eval Framework & Project Plan

**Status:** Foundation doc (2026-06-16), run as a **tracked project** (§7) ? *"Bridge the
local?Gemini rally-detection gap,"* **completed when all committed tiers land and a single quality
number is attributed (§7.1).** The empirical + research backbone for closing the gap between the
**local, no-cloud rally segmenter** and the **Gemini** path. Iterations land as their own measured
PRs, logged in [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md); this doc is the plan they execute against.

**How to read this:** §1 frames the gap with measured numbers. §2 is the precise, code-grounded
diagnosis (read before proposing fixes). §3 inventories the eval/experiment harness we already
have. §4 specifies how we strengthen that harness (the "really strong evaluation framework").
§5 is the cited external-research synthesis (4 dimensions). §6 is the prioritized roadmap with
falsifiable success criteria. **§7 is the project plan ? work-item tracker, sequencing, the single
number we attribute to the effort, and the definition of done.** §8 recaps the non-negotiable
constraints + cross-links.

> **This doc does NOT duplicate prior design.** It builds on, and cross-links:
> [`HOW_RALLY_DETECTION_WORKS.md`](HOW_RALLY_DETECTION_WORKS.md) (the 2-layer mental model),
> [`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md) (the A/B + LOVO harness, P1 shipped),
> [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) (shuttle+person+audio fusion),
> [`ANALYZERS_AND_RECONCILERS.md`](ANALYZERS_AND_RECONCILERS.md) (the signal-fusion framework +
> §13 literature review ? **already** establishes that our skeleton is standard TAL/TAS +
> late-fusion; we *cite, don't re-derive*), [`ABLATION_LAYER.md`](ABLATION_LAYER.md),
> [`EVALUATION.md`](EVALUATION.md), [`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md)
> (TAL-head-first ? VLM), and [`GOLDEN_SET_VENDORING.md`](archives/roora-vendor/GOLDEN_SET_VENDORING.md) (corpus growth).
```

---

1. The gap (measured)

Two paths produce rally segments today. **Gemini** (cloud VLM, watches the video semantically) is strong; the **local** path (WASB shuttle trajectory ? velocity windowing, no cloud) lags badly. Measured F1 vs golden labels ([`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md)):

Path	Mahadevpura (clean, single court)	Kushagra (multi-court)
Heuristic velocity windowing (local today)	0.657	**0.157**
Gen-0 owned head (LogReg on 5 trajectory features, LOVO n=6)	0.744	0.561
Two-stage WASB + Gemini refine	0.83	?
Gemini wrapper (gemini-flash)	**0.93**	**0.66**

The local heuristic is **footage-fragile**: fine on a clean single court, collapses on multi-court / busy venues. The owned Gen-0 head already closes most of the multi-court gap (0.561 vs 0.66) ? the residual is **corpus-sized, not architecture-sized**.

1.1 The 2026-06-16 local-vs-Gemini divergence run (no golden labels)

A fully-local run (`wasb_hybrid --skip-ai-handoff`) vs Gemini on 3 fresh matches (Kushagra/Yash June-12; report: `?/June 12 Local-vs-Gemini/_LOCAL_vs_GEMINI_REPORT.md`, generator `scratch/compare_local_vs_gemini.py`). **Gemini is a strong reference here, NOT ground truth** (these clips are unlabeled) ? so this is an agreement/divergence study:

	Gemini	Local (WASB no-AI)
rallies / windows (3 videos)	59	12
total "play"	6:28	**17:41**
mean window duration	**6.1 s**	**65 s** (10.6×)
true misses (gemini-only)	?	**0**
merge groups (one local window ? ?2 Gemini rallies)	?	7

Headline: **the local path does not *miss* rallies ? it fails to *segment* them.** One mahadevpura-1 window spanned the entire 174 s clip. Local "play" is 2.7× Gemini's because its mega-windows swallow the dead time between rallies.

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2. Diagnosis (code-grounded ? read before proposing fixes)

The local pipeline is **two layers** (see [`HOW_RALLY_DETECTION_WORKS.md`](HOW_RALLY_DETECTION_WORKS.md)): **(L1) detection** ? per-frame shuttle position (WASB/TrackNet); and **(L2) segmentation** ? turn the trajectory into rally intervals. The gap is almost entirely **L2**.

2.1 L1 (shuttle tracking) is NOT the bottleneck
On the divergence clips WASB reported the shuttle visible in **96?99 % of frames** (mahadevpura-1 98.6 %, GX020125 72.8 %, mahadevpura-2 78.8 %). The trajectory signal is present and dense.

2.2 L2 (segmentation) is the bottleneck ? and the flaw is conceptual
The windowing is `trajectory_to_action_windows` ([`backend/pipeline/detectors/tracknet_runner.py:256`](../backend/pipeline/detectors/tracknet_runner.py)):

- A frame is **"active"** iff the shuttle is **visible** AND its **per-second-normalized speed** exceeds `velocity_threshold` (default 0.02). Velocity = `(dist / frame_width) * (fps / dframes)` ? **already fps-normalized** (? fraction of frame-width per second).
- Active frames within `dead_time_merge_gap` (2.0 s) are **merged** into one window.
- Windows shorter than `min_action_window_duration` (2.0 s) are dropped.

The conceptual flaw: "the shuttle is moving" is treated as "a rally is in progress." During dead time the shuttle is still visible and gently moving (being picked up, walked back, knock-ups, slow serve prep) ? those frames read as *active*. The 2 s merge then bridges the brief gaps, and because (at the time this was diagnosed) there was **no `max_window_duration` cap and no inactivity/boundary split**, a single window grew until the shuttle finally went invisible ? which, at 96?99 % visibility, is rarely. Result: rally + dead-time + rally + ? collapse into a few mega-windows. *(This is the failure W1/R4 fix; both are now default-ON since the R1 milestone, PR #204 ? `max_window_duration=6`, `window_inactivity_end=1.5` in `IndexingConfig`.)*

> **Correction to an earlier hypothesis (honesty note).** A first pass guessed the cause was
> "velocity threshold not fps-invariant." It IS fps-invariant (the `fps/dframes` factor). The
> real cause is the *motion-equals-rally* conflation + 2 s bridging + no upper bound. fps matters
> only secondarily: denser 60 fps sampling keeps more slow-motion dead-time frames above the
> speed threshold. The `_LOCAL_vs_GEMINI_REPORT.md` has been corrected to match.

2.3 What this implies
The local path **has no rally-STATE signal** ? only "is the shuttle moving." Bridging the gap is fundamentally about giving L2 signals that separate *rally* from *shuttle-merely-moving* (net-crossings, two-sided play, sustained fast exchange, hit sounds) and a segmenter that produces *bounded, well-cut* intervals ? exactly the multi-signal-fusion + learned-segmenter direction the existing docs lay out. This doc makes that concrete and measurable.

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3. What we already have (eval/experiment harness inventory)

The harness is rich and composable ? we **extend** it, not reinvent it. (Modules under `backend/eval/`.)

Scoring & matching
- `metrics.py` ? temporal-IoU interval matching; precision/recall/F1, mean_iou, boundary errors.
- `segmentation_metrics.py` ? **F1@{0.1,0.25,0.5}**, **mAP@tIoU** (0.3?0.7), **segment_count_ratio** (the over/under-segmentation diagnostic). (Edit/segmental score deliberately omitted to date.)

Statistical rigor (small-n)
- `eval_stats.py` ? bootstrap 95 % CIs, paired exact **sign-test**, `paired_delta_ci`, `grouped_folds` (LOGO), `out_of_fold_predictions` (anti-stacking-leakage). The promotion bar = "?F1 CI excludes 0 **AND** sign-test agrees."
- `score_calibration.py` ? Platt + isotonic calibrators, **Brier + ECE**.

Variants, A/B, leaderboard
- `experiment.py` ? `Variant` (declarative recipe: segmenter + config + cue_flags + gates + learned-classifier + threshold/merge_gap/calibrate), `compare()` (LOVO-honest A/B with an "honest" stats block), `compare_unlabeled()` (agreement on unlabeled footage), `CONTROL` (pinned baseline), `record_experiment()` ? `eval_baselines/experiment_leaderboard.json`.
- `ablation.py` ? leave-one-signal-out marginal ?F1 with CI + sign-test + a no-over-claim verdict vocabulary (`earns_keep` / `suggestive` / `inert` / `structural_protected`).
- `classifier.py` ? dependency-free numpy `LogisticRegression` (the learned scorer; JSON-serializable).

Eval?serve discipline + gates
- `windowing.py` ? single-sources the **SERVED** preset from `IndexingConfig` defaults so eval and production can't silently diverge; `PROPOSAL` (recall-oriented) is where the golden feature substrate is built. (This exists *because* Gate-A measured **+0.074 on PROPOSAL but ?0.008 on SERVED** and was reverted ? the canonical eval?serve lesson.)
- `promotion.py` / `served_gate_a.py` ? promote a cue only if it does NOT regress at the served windowing; `per_user_gate.py` / `per_user_regression.py` ? per-user analogues; `regression.py` ? golden-set no-regression baseline with a GT-hash to detect label changes.

Golden substrate
- `gt_loader.py` ? one canonical GT loader (both legacy formats). `fusion_golden.py` / `fusion_compare.py` / `fusion_audio.py` ? extract replayable per-window feature JSONs (shuttle 5 + person 5 + audio 3) so ablation/compare re-score on CPU without re-detecting.
- **Corpus today: 15 golden videos** (started at the 6-video label-set hash `50d98ffbc370`, since grown with the GX/Boxhill multicourt clips); more expected via the Roora vendoring track ([`GOLDEN_SET_VENDORING.md`](archives/roora-vendor/GOLDEN_SET_VENDORING.md)).

Signals available to L2 (built, default-OFF): net-crossing count (`rally_gate.py`, Gate-A), 5 person/court features (`person_detector.py` + `fusion_features.py`), 3 audio-hit features (`audio_features.py`). Fusion hooks: `fusion_hybrid.py::_enrich_candidate_stats` / `_select_for_handoff`. Persisted in `candidate_segments.stats`.

3.1 Known gaps in the harness (what §4 fills)
1. **Over-segmentation is under-surfaced in the default report.** `segment_count_ratio` exists but the merge/split *breakdown* (the very thing that exposed the over-merge) lives only in a scratch script. tIoU-F1 alone is **blind to over-merge** ? a giant window can still "match" via overlap.
2. **No segmental edit-score** (Levenshtein over the predicted/gt label sequence) ? the standard TAS over-segmentation metric.
3. **Per-cue calibration is fitted but never *measured* post-hoc** (no ECE/Brier delta reported).
4. **`out_of_fold_predictions` (anti-leakage stacking) is built but not wired to an arbiter.**
5. **No active-learning loop** ? nothing tells us *which* unlabeled video to label next.
6. **No single "rally-segmentation eval" entrypoint** ? the pieces exist but a one-command `score this segmenter vs golden, honestly` runner would make every iteration cheap.

7. **The local-vs-Gemini agreement track is ad-hoc** (a scratch script), not a first-class
   shadow signal alongside the golden gate.

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4. The strengthened evaluation framework (the "really strong harness")

Design principles (inherited, non-negotiable):
- **Signals ? learned scorer, NO special-casing** ([`ANALYZERS_AND_RECONCILERS.md`](ANALYZERS_AND_RECONCILERS.md) §1):
  every new cue is a feature column the scorer learns to weight; never an `if/else` in the decision path.
- **Promote only on measured lift; default OFF; revert is one flag** ([`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md) §6).
- **eval ? serve:** a harness win must be re-confirmed on the *served* windowing before any default flip.
- **Honest small-n stats:** never a bare mean at n?6?16; always CI + paired sign-test (C1).

4.1 Metric set the leaderboard should standardize on
Surface ALL of these in the default `compare()` report (most already computed; the task is to
report them together so over-merge can never hide again):

- **tIoU-F1 @0.5** (primary) **and F1@{0.1,0.25,0.5}** (segmental, TAS-standard).
- **mAP@tIoU** (0.3?0.7) when predictions carry confidences.
- **`segment_count_ratio`** + a **merge-rate / split-rate** breakdown (fraction of GT rallies that
  share a predicted window = under-segmentation; fraction of predicted windows that span ?2 GT =
  the over-merge signal). Promote the scratch `compare_local_vs_gemini.py` matching into a reusable
  `backend/eval/` helper.
- **Segmental edit (Levenshtein) score** ? the canonical over-segmentation metric we currently lack.
- **Boundary timing** (mean start/end error on matched pairs) ? already in `metrics.py`, surface it.
- **Calibration (ECE/Brier)** reported pre/post per-cue calibration (close gap #3).

> **Why:** plain tIoU-F1 rewarded our mega-windows (they "overlap" real rallies). The metric set
> above makes over/under-segmentation *first-class*, so a fix that improves segmentation is
> visible even when tIoU-F1 barely moves ? and vice-versa.

4.2 Splits & significance
- **Leave-one-video-out, grouped by match** (never leave-one-*window*-out ? windows within a video
  are correlated). Keep `grouped_folds`. Note the **RLOOCV bias** caveat for tiny n (see §5.4).
- **Bootstrap CI + paired exact sign-test** on per-video held-out F1 (C1). A result is *promotable*
  only if CI excludes 0 **and** the sign-test agrees, **and** it survives the served-windowing check.
- **Anti-leakage:** per-LOVO-fold threshold tuning + calibration on the TRAIN fold only (already
  enforced in `experiment._calibrate_and_tune`); wire `out_of_fold_predictions` once an arbiter exists.

4.3 The golden corpus + active learning (the highest-leverage lever)
The single biggest quality lever is **more labeled matches** ([`NEXT_STEPS.md`](NEXT_STEPS.md):
"corpus growth = highest leverage"). With ~6 ? ~16 videos:
- Grow via the Roora vendoring flow ([`GOLDEN_SET_VENDORING.md`](archives/roora-

vendor/GOLDEN_SET_VENDORING.md),
[`ROORA_LABELING_GUIDELINES.md`](archives/roora-vendor/ROORA_LABELING_GUIDELINES.md)) ? court calibration, IAA, the
3 non-negotiables (ignore dead time, every rally, one-contact = one shot).
- **Active learning:** stop labeling videos at random. Score the *unlabeled* corpus and label the videos where the local segmenter is most *uncertain* / most *divergent from Gemini* / most *diverse* (new venue, fps, lighting). §5.3 surveys the methods; the local-vs-Gemini agreement signal (§4.4) is a ready proxy for "where do we disagree most ? label that."

4.4 Shadow evaluation: Gemini-agreement as a free, unlimited signal (NOT a gate, NOT training data)
We have exactly one strong reference available on *unlabeled* footage at inference time: **Gemini**.
- `experiment.compare_unlabeled()` already scores agreement (agreed / a-only / b-only) without labels.
- Use it as a **shadow metric**: run any candidate segmenter on a large unlabeled pool, measure agreement-with-Gemini, and watch it as a *trend* ? cheap, unlimited, catches regressions the 6 golden videos can't. **It is a shadow signal only.** Golden labels remain the gate; Gemini is not ground truth (it has its own errors) and ? per the licensing guardrail (§8) ? **its outputs may never become training data for owned weights.**

4.5 One-command rally-segmentation eval (close gap #6)
Wrap the above into a single reusable entrypoint (a future PR ? not built here): given a segmenter variant + the golden manifest, emit the full §4.1 metric set + the honest stats block + a leaderboard row, and (optionally) the §4.4 shadow agreement on an unlabeled pool. Every iteration in §6 is then "run the entrypoint, read the honest delta, promote-or-not."

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5. External research synthesis (cited, adversarially verified)

> Produced by a deep-research run (6 search angles ? 27 sources ? 126 candidate claims ? 25
> 3-vote-adversarially-verified, 23 confirmed / 2 refuted). **Cite, don't over-claim.** Two claims
> were *killed* in verification ? recorded in §5.5 so we don't repeat them. **License caveat:** the
> code repos below are research releases; **independently confirm each repo's actual license is
> MIT/Apache-2.0/BSD before vendoring any of it** (this survey did not verify every license file).
> **Governance:** none of these techniques require training on paid-LLM outputs ? all are compatible
> with our no-Gemini-training-data guardrail (§8).

5.1 Windowing & boundaries (TAL/TAS) ? the explicit-boundary machinery we lack
The recurring lesson: modern action-localization models have an **explicit boundary mechanism**; our heuristic has none (it merges on a velocity+gap rule). The transferable *idea* is "predict a boundary and cut there," but the deep models are built for **hundreds+ of training videos**, so at n=6?16 they are **research bets**, not drop-ins.

- **ASRF ? Action Segment Refinement Framework** (Ishikawa et al., WACV 2021, [arXiv:2007.06866](https://arxiv.org/abs/2007.06866)). A shared feature extractor feeds **two branches**: frame-wise classification + a **Boundary Regression Branch**; predicted boundaries refine/split the segmentation (+13.7% edit, +16.1% segmental F1 over MS-TCN-era SOTA on 50Salads/GTEA/Breakfast). **Why it fits:** the boundary head is exactly the missing "cut here" mechanism. **? Two caveats:** (1) ASRF's headline goal is to *reduce* over-segmentation (merge spurious fragments) ? the opposite of our over-*merge*; only the *split-at-a-detected-boundary* sub-mechanism transfers. (2) The claim that ASRF's boundary branch is a **model-independent post-hoc splitter** you can bolt onto any dense signal was **REFUTED** (§5.5) ? you must train the two-branch model, not graft the branch. **Effort/payoff:** research bet.

- **TriDet** (Shi et al., CVPR 2023, [arXiv:2303.07347](https://arxiv.org/abs/2303.07347)). A **Trident-head** models each boundary as a *relative probability distribution over local bins* (boundary = expected value over neighboring instants) ? built for imprecise/ambiguous boundaries. 69.3% avg mAP on THUMOS14 (> ActionFormer 66.8% at ~75% latency), **convolution-only (SGP), no self-attention** (attractive on a single local GPU). Ablation shows the Trident-head's gain concentrates at strict tIoU=0.7 ? it buys *boundary precision*. **Effort/payoff:** research bet (trained on 100s of videos); the portable piece is the relative-boundary idea.

- **ActionFormer** (ECCV 2022, [arXiv:2202.07925](https://arxiv.org/abs/2202.07925)). The reference single-stage anchor-free template: multiscale pyramid + local self-attention; classify every moment + regress its boundaries (71.0% mAP@0.5 THUMOS14). This is the family TriDet/TemporalMaxer/CausalTAD extend, and the natural target if we ever graduate from heuristic+LogReg. **Research bet.**

- **TemporalMaxer** (CVPRW 2023, [arXiv:2303.09055](https://arxiv.org/abs/2303.09055)) ? **the most important small-data signal in this dimension.** It replaces ActionFormer's self-attention block with a **parameter-free max-pool** and *beats* it (67.7 vs 66.8 avg mAP) at **~4× fewer params**. The ablation is the takeaway: a **learned 1-D conv block is the WORST** (59.4, overfits redundant features); parameter-free pooling wins. **Why it fits:** direct evidence to **keep our scorer parameter-light** (the numpy LogisticRegression over a handful of features is the *right* choice at our scale, not a placeholder to rush past). **Effort/payoff:** a discipline (free) ? adopt the principle, not the model. *(Caveat: the overfitting evidence is about feature redundancy on a mid-size set, a directional inference to our small-n setting.)*

- **CausalTAD** (2024, [arXiv:2407.17792](https://arxiv.org/abs/2407.17792)) ? *steering-away note.* Causal-attention masking (attend only to past/present) won EPIC-Kitchens/Ego4D 2024 tracks, but it adds **no** boundary-split module. Useful later for *online/streaming* rally detection; **NOT** the over-merge fix. Listed so we don't mistake causal attention for the remedy.

- **Metrics (the actionable cross-cut):** the field leads with **segmental Edit (Levenshtein) score and F1@k**, not frame accuracy (MoF) or plain tIoU ? see §5.4.

5.2 Shuttle-trajectory ? rally signal processing ? published badminton systems already do what we lack

This is the highest-confidence, lowest-effort dimension: in-domain papers operationalize the exact boundary heuristics we're missing, in spirit license-compatible (re-confirm before vendoring code).

- **See et al. ? Service & End-of-Rally Detection in Badminton** (2024 Springer chapter [10.1007/978-3-031-69073-0_15](https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15); 2025 SN Computer Science follow-on [10.1007/s42979-025-03935-0](https://link.springer.com/article/10.1007/s42979-025-03935-0)). **(a) Rally END = a sustained-inactivity rule:** the rally ends when **?80% of consecutive frames show no shuttle motion** (~**95% accuracy** over 187 rallies). **(b) Rally START (service) = a learned classifier** over court geometry + player poses (~26 features/frame × 12 frames ? CNN/BiLSTM, **88.6%** accuracy). **Why it fits:** the inactivity rule is *precisely* the end-of-rally cut our windowing lacks ? a few lines, directly breaks 65 s mega-windows. **? Caveats:** the rule is a *termination*, not a *within-rally split*; results rest on paywalled abstracts + small test sets (35?187 rallies); the serve classifier needs the (default-OFF) person cue + court polygons. **Effort/payoff:** the inactivity rule = **low-effort high-confidence win**; serve classifier = medium.

- **Hsu, Yu & Cheng ? trajectory + swing fusion** (Sensors 2024, 24(13):4372, [10.3390/s24134372](https://www.mdpi.com/1424-8220/24/13/4372)). Fusing TrackNet trajectory with a YOLOv7 swing detector via a Shot Refinement Algorithm raises hit detection **F1 72.3% ? 90.5%** (over 69 BWF matches / 1582 rallies). Hits are found by **trajectory direction-reversal / peak detection** (convert (x,y)?(frame,y) wave peaks) then reconciled with the swing cue. **Why it fits:** validates our **late/score-level fusion** approach in-domain, and direction-reversal/peak is a cheap **per-window feature** marking contact events (a "rally has hit cadence" signal ? the discriminator a "shuttle-moving" heuristic lacks). **? Important caveat (the one 2-1 vote):** this paper does *within-rally stroke/hit* segmentation, **not** rally start/end ? "it splits merged rallies" is *our extrapolation*; frame it as "hit cadence plausibly helps locate rally boundaries," not a proven rally-splitter. **Effort/payoff:** the peak/direction-reversal feature = low-effort win (feed the scorer); full swing fusion = medium.

- **What best separates "rally" from "shuttle merely moving"** (synthesis): sustained-inactivity termination (See), hit cadence / direction-reversals (Hsu), net-crossings ?2 (our Gate-A, [`rally_gate.py`](../backend/pipeline/detectors/rally_gate.py)), two-sided court occupancy (our person features). None alone is sufficient; the spine (§4) says **feed them all to the learned scorer**.

5.3 Small-data strategies ? most quality per label

- **Timestamp / sparse supervision** (Li et al., CVPR 2021, [arXiv:2103.06669](https://arxiv.org/abs/2103.06669); TAS survey Ding et al., [arXiv:2210.10352](https://arxiv.org/pdf/2210.10352)). Label **one frame per action** ? iteratively refine pseudo frame-labels ? *comparable* to full supervision (50Salads: timestamp 75.6% acc / 60.1 F1@50 vs full 83.7% / 70.1). **Why it fits:** sparse per-rally timestamps cost far less than dense boundary labels ? relevant if we ever train a frame-wise model. **? Caveats:** "comparable" hides a real **~8?10 pt residual gap**; annotation savings are modest (~35%, annotators still watch the whole clip); benchmarks are cooking/egocentric ? transfer to 1-D shuttle trajectories is **unproven**. **Effort/payoff:** research bet to pilot.
- **Parameter-light scorer** ? see TemporalMaxer (§5.1): keep the numpy LogisticRegression; resist a learned temporal net until the corpus is much larger. **Low-effort win (a discipline).**
- **? OPEN (the research could NOT verify these ? honest gap):** *active learning* (which video/segment to label next via uncertainty/diversity/core-set) and the *teacher-student / distillation governance boundary* for using Gemini as a weak/pseudo-labeler returned **no surviving verified claim**. Our pragmatic, guardrail-safe stance (not from the literature ? from our own design): pick the next video to label by **maximal local-vs-Gemini disagreement** (the §4.4 shadow signal) + **scorer uncertainty** + **venue/fps diversity**; and treat **Gemini strictly as an inference-time weak-reference / eval oracle ? never a training-data source for owned weights** (§8). These two topics deserve their own grounded research pass before we lean on them.

5.4 Evaluation rigor at small n ? lead with segmental metrics

- **Frame accuracy (MoF) and plain tIoU are BLIND to over-segmentation** (TAS survey, Ding et al., [arXiv:2210.10352](https://arxiv.org/pdf/2210.10352); segmental-F1 origin Lea et al., CVPR 2017). "The score can be high even when segments are fragmented." The over-merge that bit us is exactly what tIoU-F1 hides (a giant window still *overlaps* real rallies). **Action:** lead the leaderboard with the **segmental Edit (Levenshtein) score** (penalizes spurious ordering/fragmentation) **and F1@{0.1,0.25,0.5}**, paired with our existing `segment_count_ratio` + the merge/split breakdown. **We already have** F1@k, mAP@tIoU, segment-count-ratio, bootstrap CIs, paired sign-test, Platt/isotonic + ECE/Brier ? the only *missing* primary metric is the **Edit score** (low-effort add).
- **? OPEN (not covered by a verified claim):** IAA / label-noise level in the golden set, and which over-segmentation-robust loss (ASRF boundary/smoothing, MS-TCN truncated-MSE) best tolerates it at small n. Measure IAA as the corpus grows (Roora has an A/A overlap plan); revisit robust losses only when a frame-wise model is on the table.

5.5 Refuted claims (recorded so we do NOT repeat them)
1. **"ASRF's boundary branch is model-independent ? a post-hoc splitter applicable to any dense segmentation output."** **REFUTED (0-3).** Implication: we cannot bolt ASRF's boundary head onto our trajectory signal as a drop-in; the two-branch model must be trained together. ? port the *concept*, don't expect a plug-in.
2. **"A GFL-inspired Boundary Distribution Regression head is the mechanism most relevant to splitting merged windows."** **REFUTED (0-3)** (and the cited source had a suspect identifier). Do not adopt this framing as established.

5.6 External due-diligence review (2026-06-17) ? owner-commissioned; verify before submission
An owner-commissioned external review (31 cited sources) of the [`RALLY_DETECTION_QUALITY_REPORT.md`](RALLY_DETECTION_QUALITY_REPORT.md) checkpoint **validated the methodology** (the decoupled 2-layer design, the R2 metric, LOVO + paired-sign-test discipline, the consented-data + over-seg-aware-harness moat) and surfaced these **borrowable inputs**. These are *research inputs for later pickup* ? they add **no measured numbers**, and the citations are the review's own and **must be re-verified before any submission**.

- **Positioning ? the "first-mile" gap (highest-value).** Badminton stroke-classification SOTA (**BST** ? Badminton Stroke-type Transformer, CVPR-W 2026; **DSTA** ? Decoupling Spatio-Temporal Adapter) and the big datasets (**ShuttleSet** ~36,492 strokes; **Fine-Badminton** ~27,597 actions) all **silently assume a pre-trimmed rally clip already exists** ? they depend on a human having segmented the untrimmed match. Our system solves that ignored *prerequisite*. Reframe the paper as **"the missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage"** ? a real, unoccupied gap and a far stronger narrative than "applied/systems."
- **R2 ? Action-Spotting / Precise-Event-Spotting (AS/PES).** Our asymmetric tolerance-match metric sits at the **TAL × AS intersection** (interval task, tolerance-based match). Grounding it in the AS/PES literature (SoccerNet, TenniSet, T-DEED, the PES survey [arXiv:2505.03991]) frames R2 as a deliberate methodological contribution, not an ad-hoc metric ? strengthens §5.4 and the report's §9.
- **Comparative baselines reviewers will require.** Benchmark against **MonoTrack** and **BST** on **ShuttleSet / Fine-Badminton** ? we currently have only a generic heuristic + the Gemini wrapper; domain-specific baselines are a hard publication requirement.
- **Scorer upgrade ? prefer TemporalMaxer.** For the post-logreg scorer (corpus-gated; NEXT_STEPS M0.4), **TemporalMaxer** (parameter-free temporal max-pool; ~2.8× fewer GMACs; robust at small-N, [arXiv:2303.09055]) fits the cheap/local/fast mandate better than attention-heavy ActionFormer.
- **Court polygons ? pose ? the rally-END lever.** Court masking is not only the person-cue spectator fix (§4 / `MULTI_SIGNAL_FUSION_PLAN.md`): once pose is isolated to on-court players it can resolve **ambiguous, occluded end-of-rally states** ? feeding directly into the **#1 remaining gap (the

rally-END boundary)**, not just precision. Connect court-polygons to the R4/END-trim work.
- **Name the eval?serve lesson "pipeline distribution skew."** Downstream-cue value depends on the upstream proposal distribution (the Gate-A episode) ? a citable framing and candidate paper "lesson."
- **Citable Layer-1 anchor (public benchmark, NOT our footage).** WASB SBDT F1 95.6 (step=1) / 94.0 (step=3) vs MonoTrack 92.1, TrackNetV2 89.4, BallSeg 71.7 (WASB, BMVC 2023). Justifies the frozen detector choice; **keep distinct from our own, still-unmeasured footage domain-gap** (§2.1, a real open limitation the review also flags).

Sequencing: these are *publication-prep + post-current-idea* levers. The prerequisites remain corpus growth to n?16 and **multi-court isolated as its own reported slice** (§6 Tier-3 / §7) ? do those first; the items above land once current windowing/fusion ideas are exhausted.

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6. Prioritized roadmap (cheap ? durable)

Every item is **its own measured PR**: default-OFF, scored on the §4 metric set over the golden set with honest stats (CI + sign-test), **re-confirmed on the served windowing** (the eval?serve gate), logged in [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md). Ordered by the research's effort/confidence tiering. Each carries a **falsifiable success criterion** ? if it isn't met, the item does not promote (and that null is itself recorded).

Definition of "bridging the gap"
Local rally segmentation, measured **honestly LOVO-by-match on the golden set**, approaches the Gemini reference (today **0.93 single-court / 0.66 multi-court** tIoU-F1) ? but tracked primarily through the **over-segmentation-sensitive metrics** (segmental Edit + F1@k + low merge/split rate), since those, not bare tIoU-F1, capture our actual failure. A secondary, free signal is rising **agreement-with-Gemini** on the unlabeled pool (§4.4, shadow only).

Tier 0 ? Make the failure measurable FIRST (do before any fix)
- **E1 · Over-segmentation-aware leaderboard.** Add the **segmental Edit (Levenshtein) score** to `segmentation_metrics.py`; surface Edit + F1@{0.1,0.25,0.5} + `segment_count_ratio` + a **merge-rate / split-rate** breakdown (promote the `scratch/compare_local_vs_gemini.py` matching into a reusable `backend/eval/` helper) in the default `experiment.compare()` report; add the one-command rally-segmentation eval entrypoint (§4.5). **Success:** the current heuristic's over-merge shows up as a *number* (low Edit, high merge-rate) on the 6 golden videos, and a baseline row lands in the leaderboard. *(Low effort, high confidence ? research-backed §5.4.)*

Tier 1 ? Low-effort, high-confidence wins
- **W1 · Bounded windowing (the direct over-merge fix).** Add a `max_window_duration` cap and a **sustained-inactivity split** to `trajectory_to_action_windows` ? port See et al.'s rule (split /

end a window when ?~80% of consecutive frames over a trailing window show no shuttle motion). Config-gated, default preserves current behavior. **Success:** segmental Edit + F1@k improve vs `CONTROL` on the golden set (?F1 CI excludes 0 **and** sign-test agrees) with **no served-windowing regression**; merge-rate drops materially. *(Win ? §5.2 See et al., ~95% end-of-rally acc.)*
- **W2 · Rally-state features into the learned scorer (no special-casing).** Wire the already-built, default-OFF cues as scorer feature columns: **net-crossings** (Gate-A), **two-sided motion**, and a **new cheap trajectory feature ? direction-reversal / peak count** (Hsu) as a hit-cadence proxy; audio hit-rate when the hit detector lands. **Re-measure with W1 in place and on the served windowing** ? recall person was ?0.021 and audio +0.079 but *both CIs spanned 0* at n=6 on the *pre-fix over-merged* candidates; the picture may change once windows are bounded. **Success:** the ablation (`ablation.py`) shows a cue with ?F1 CI clearing 0 + sign-test, surviving served re-check ? promote that cue; others stay default-OFF + retained (no deletion on a small-n null). *(Win/medium.)*
- **W3 · Keep the scorer parameter-light (a discipline, documented not coded).** Per TemporalMaxer's ablation (learned temporal conv overfits; parameter-free wins), resist replacing the numpy LogisticRegression with a learned temporal net until the corpus is much larger. Record this as a standing decision. *(Win ? §5.1 TemporalMaxer.)*

Tier 2 ? Medium effort
- **M1 · Serve-start classifier ? rally-start boundary** (See et al. pose/court, 88.6%). Needs the person cue + a **court-mask source** (manual polygon vs auto-homography ? *owner-gated* per [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) §9.1). **Success:** start-boundary timing error drops on golden; promotes only if it lifts segmental F1@k without served regression.
- **M2 · Per-venue threshold calibration** against golden labels (the heuristic's footage-fragility ? it scored 0.157 on multi-court). Tune `velocity_threshold` + `merge_gap` so local segment counts track golden counts per venue. **Success:** multi-court F1 rises toward the Gen-0 head's 0.561 without harming clean-court F1. *(Velocity is already fps-normalized ? calibrate per-venue, not per-fps.)*

Tier 3 ? Research bets (durable; corpus-gated)
- **R1 · Owned segmenter with an explicit boundary branch** (the ASRF/TriDet *idea* ? a boundary head that predicts cut points ? ported into our owned TAL head, NOT the full deep model, and **trained**, not grafted (§5.5 refutation)). Per [`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md): **TAL head first, VLM later.** **Gate:** the central unverified assumption is whether boundary-head gains hold at our scale on trajectory features ? so **pilot only when the corpus is materially larger (~?16?30 matches)**, and only after W1/W2 establish the heuristic ceiling. **Success:** beats the tuned heuristic + LogReg on held-out matches (segmental Edit + F1@k),

honestly LOVO.
- **R2 · Timestamp/sparse supervision** to get more labeled rallies per annotator-hour (§5.3) ? pilot
  if/when we train a frame-wise model; research bet (residual gap, unproven on trajectories).

Throughout ? corpus growth + active learning + label quality
- **Grow the golden set 6 ? ~16** via Roora ([`GOLDEN_SET_VENDORING.md`](archives/roora-vendor/GOLDEN_SET_VENDORING.md)) ?
  the single highest-leverage lever ([`NEXT_STEPS.md`](NEXT_STEPS.md)).
- **Active learning:** select the next videos to label by max **local-vs-Gemini disagreement** (§4.4)
  + scorer uncertainty + venue/fps diversity (our pragmatic stance; §5.3 flags the formal methods as
  an open research pass).
- **Measure IAA / label noise** as the corpus grows (Roora A/A overlap); revisit over-segmentation-
  robust losses only when a frame-wise model is on the table (§5.4 open question).

Dead-ends / do-NOT (so we don't relitigate)
- Don't treat **causal attention** (CausalTAD) as the over-merge fix ? it has no boundary-split (§5.1).
- Don't expect **ASRF's boundary branch as a post-hoc drop-in** on our dense signal ? refuted (§5.5).
- Don't adopt a **learned temporal net** at n=6?16 ? TemporalMaxer evidence says it overfits (§5.1).
- Don't chase **bare tIoU-F1 / frame accuracy** ? blind to the over-merge (§5.4).
- Don't let **Gemini outputs become training data** for owned weights ? inference/shadow only (§8).

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7. Project: plan, tracking & definition of done

This roadmap is run as **one tracked project ? "Bridge the local?Gemini rally-detection gap."** It
completes when all committed tiers land AND a single quality number is attributed to the effort,
measured honestly and logged. This §7 is the living burn-down (update the status column as PRs merge;
log each result in [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md), numbers in
`eval_baselines/experiment_leaderboard.json`).

7.1 The number (what we attribute to this effort)
Measured **leave-one-video-out, grouped by match, on the golden set** (grown to ?~16):
- **Headline:** multi-court rally-segmentation **tIoU-F1**, `baseline ? final`, plus **gap-closure %**
  = `(final ? baseline) / (Gemini ? baseline)`. Today: heuristic **0.157** ? Gemini **0.66**
  (Gen-0 head already **0.561**). **Success bar (proposed, owner-adjustable): close ? 50% of the
  heuristic?Gemini multi-court gap** ? i.e. local multi-court F1 ? ~0.41, ideally beating Gen-0's 0.561.
- **Primary lens (over-segmentation-sensitive, the actual failure):** segmental **F1@0.25** +
  Edit-score + **merge-rate** ? these must improve even if tIoU-F1 moves little.
- **Reported with honest stats** (bootstrap CI + paired sign-test); a result counts only if it also
  holds on the **served** windowing (eval?serve gate). The **baseline is frozen at E1** before any fix.
- **Tracked headline = the n=6 golden *aggregate* tIoU-F1@0.5** (LOVO; the robust unit, not a single
  video) via `python -m backend.eval.rally_seg_eval`. The 0.157/0.66 figures above are per-venue
  references; computing an exact "% gap-to-Gemini closed" needs Gemini measured on the

same golden set + metric ? **TODO** (don't mix the per-venue refs with the aggregate).

Progress (cumulative, golden n=6 aggregate tIoU-F1@0.5):

after	F1	merge_rate	? vs baseline
E1 baseline (current heuristic)	0.300	0.523	?
+ W1 bounded windowing (cap=12, **default-OFF**)	**0.439**	**0.038**	**+0.139** (6/0 sign-test)
+ W2 net-crossings gate (mc=1, **default-OFF**)	**0.457**	0.038	**+0.157 cumulative** (W2 step +0.019, 6/0)
+ W1.5 hard-cap fallback (**default-OFF**)	0.444	0.000	**+0.006 (n.s., CI spans 0)** ? neutral on golden; real-footage fix (mahadevpura-1 174s blob ? 24×?9s windows)
+ W2 multi-feature scorer (person/audio)	_next (GPU)_		

7.2 Work items (status checklist)
Status: ? not started · ? in progress · ? done. (Each row = its own default-OFF, measured PR.)

ID	Deliverable	Tier	Depends on	Acceptance / promote criterion	Status
E1	merge/split-rate metric + F1@k surfaced + one-command `rally_seg_eval` entrypoint; **froze baseline** (#185)	0	?	over-merge shows up as a number on the 6 golden videos	? baseline F1 0.300, merge_rate 0.523
W1	Bounded windowing ? `max_window_duration` cap + inactivity split (See et al.) (#187)	1	E1	seg F1@k ? vs baseline (CI?0 **and** sign-test), no served regression; merge-rate ?	? **?F1 +0.139 (6/0), merge_rate 0.523?0.038** (default-OFF)
W1.5	Hard-cap fallback (`window_hard_cap`) ? chop continuously-active windows at the cap when no inactivity gap exists	1	W1	golden ?F1 ?0 (or neutral) **+** fixes a real-footage over-merge it can't otherwise cap	? **golden ?F1 +0.006 (n.s.)**; real `mahadevpura-1` 174s blob ? 24×?9s; retained **default-OFF** (real-footage + true-cap rationale, not a golden lift)
W2	Rally-state features (net-crossings ? #188 + **never-empty safeguard** ?; two-sided motion, **direction-reversal/peak**, audio = follow-on)	1	E1, W1	?1 cue ?F1 CI clears 0 + sign-test, survives served re-check ? promote; others retained default-OFF	? **net-crossings ?F1 +0.019 (6/0)**, gate now **do-no-harm** (never zeroes a video); multi-feature scorer next (GPU)
W3	Keep scorer parameter-light (standing decision; TemporalMaxer evidence)	1	?	documented; no learned temporal net until corpus ? now	?
M1	Serve-start classifier ? rally-start boundary (pose/court)	2	W1, **court-mask decision (owner-gated)**	start-boundary timing error ?; promote on seg-F1 lift, no served regression	?
M2	Per-venue threshold calibration vs golden	2	E1	multi-court F1 ? toward Gen-0 0.561 without harming clean-court	?
R1	Owned segmenter with an explicit **boundary head** (port ASRF/TriDet *idea*, trained ? not grafted)	3 (stretch)	corpus ?~16?30, W1/W2	beats tuned heuristic+LogReg on held-out matches (seg Edit + F1@k), LOVO	?
R2	Timestamp/sparse-supervision pilot (label-efficiency)	3 (stretch)	R1 path	measured label-efficiency win	?
C	Golden corpus 6 ? ~16 (Roora) + active-learning selection + measure IAA	ongoing	?	corpus ?16; IAA recorded	? (~6 now; ~10 by the weekend)

7.3 Sequencing
`E1` (metrics + baseline) ? `W1`/`W2`/`W3` (the high-confidence wins) ? `M1`/`M2` ? `R1`/`R2` (corpus-gated). `C` (corpus growth) runs in parallel throughout ? it is the single highest-leverage input and gates the Tier-3 bets.

7.4 Definition of done
The project is **complete** when:
1. **Tier 0 + Tier 1 + Tier 2** work items are merged ? each a measured PR (default-OFF, promoted only on measured lift, re-confirmed on the served windowing).

557 2. **The headline number (§7.1) is recorded** in `QUALITY_ITERATIONS.md` + the leaderboard, with honest
558 stats, on the grown golden set ? **and the gap-closure target is met, or the honest null is recorded
559 with a reasoned go/no-go on Tier 3.**
560 3. Each landed item left the harness greener (no regression) and `QUALITY_ITERATIONS.md` tells the story.
561
562 > **Tier-3 scope decision (owner-adjustable):** R1/R2 (the owned boundary-head model + sparse
563 > supervision) are **corpus-gated stretch goals** ? the verified research says deep TAL detectors risk
564 > overfitting at n=6?16. **Default:** the project *completes* at Tier 0?2 + an attributed number, and
565 > Tier 3 spins out into the owned-model milestone ([`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md)).
566 > Fold Tier 3 into the DoD instead if you want the project to span the owned model.
567
568 ---
569
570 ## 8. Constraints & guardrails (non-negotiable) + cross-links
571
572 - **Licensing:** every dependency ? code, data, AND weights ? must be **MIT / Apache-2.0 / BSD**.
573 Reject viral/NC/custom (Gemma-3, Llama-4 MAU cap, Commons-Clause).
574 - **Never train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs** (terms/copyright);
575 paid LLMs are inference/serving/shadow-reference only. Also forbidden: public datasets known to
576 be LLM-generated. **Exclude minors** from any training pool; per-user data needs explicit opt-in.
577 - **Owned model:** TAL head FIRST (de-risked, trains in minutes, doubles as a pseudo-label teacher +
578 honest baseline the VLM must beat), then Qwen3-VL (Apache-2.0) / InternVL3 (MIT) ? see
579 [`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md).
580 - **Cross-links:** [`HOW_RALLY_DETECTION_WORKS.md`](HOW_RALLY_DETECTION_WORKS.md) ·
581 [`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md) · [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) ·
582 [`ANALYZERS_AND_RECONCILERS.md`](ANALYZERS_AND_RECONCILERS.md) · [`ABLATION_LAYER.md`](ABLATION_LAYER.md) ·
583 [`EVALUATION.md`](EVALUATION.md) · [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md) ·
584 [`GOLDEN_SET_VENDORING.md`](archives/roora-vendor/GOLDEN_SET_VENDORING.md) · [`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md).
585

8. assistant (2026-06-21T13:56:27.855Z)

I have the core doc. The §7 tracker is rich. Let me read the remaining track docs to confirm status and capture details, especially R2/R4, Gate-A wiring, and the experiment harness.

9. user (2026-06-21T13:56:28.761Z)

1 # R2 ? Task-faithful rally-boundary eval metric (tolerance-match + over-seg guard)
2
3 **Status:** **IMPLEMENTED (augmenting), 2026-06-17** ? owner-reviewed + decisions locked 2026-06-17;
4 `backend/eval/tolerance_metrics.py` + `rally_seg_eval` reporting landed (default-additive; tIoU stays the
5 gate pending the ablation experiment). Gates the rally-end work (R4) and the dual-eval-set regression gate.
6 A focused increment of the rally-quality project (RALLY_QUALITY_RESEARCH.md §7;
7 the day-1 validation that motivates it: REAL_FOOTAGE_VALIDATION_REPORT.md;
8 the directive: `eval-boundary-tolerance` + `eval-dual-set-regression`).

> **Implementation refinement (2026-06-17):** §2.3.2's boundary-error distribution is computed over
> **unconditional best-overlap** matches, NOT the within-? 1:1 matches ? a ?-gated set is survivorship-biased
> (loose ends fail the gate and vanish, hiding the very end-deficit the diagnostic exists to surface; that
> within-? deficit instead shows up as low `tol_recall`). Matches the day-1 validation method. Measured on the
> n=9 corpus: **|?start| 1.25 s vs |?end| 2.19 s** ? "start?solved, end=the gap", confirmed.

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0. TL;DR
`tIoU-F1@0.5` is the wrong ruler for rally detection, in **both** directions: it over-penalizes the
intrinsically fuzzy rally START (the ~1?1.5 s service window), and its overlap matching is murky about
over-production. R2 replaces the headline with a **tolerance-match metric under strict 1:1 (Hungarian)
assignment**, reported as a **decomposition** ? `P@? / R@? / F1@?` + **boundary-error distributions split
into start/end** + an **over-segmentation guard** + a **?-sweep** ? not a single number.

Honest, slightly uncomfortable consequence (measured, see §3): under R2 the local detector's headline
number *drops* and the gap to Gemini *widens* vs what tIoU showed ? because R2 faithfully charges for the
1.5× over-production and the loose ends that tIoU's overlap matching partly absorbed. That is the point:
R2 splits "behind Gemini" into its two real causes ? **(a) over-production [precision]** and **(b) loose
rally-ends [the R4 lever]** ? and forgives only the inherent start fuzziness. It is a more honest ruler,
not a flattering one.

---

1. Why `tIoU@0.5` is the wrong ruler (grounded in the n=2 ground-truth clips)
1. **Over-penalizes the START.** For a 5.7 s rally, `tIoU ? 0.5` tolerates only ~±1.9 s of boundary shift
   (`? = D·(1??)/(1+?)`). The service window (stance ? racquet contact) alone is ~1?1.5 s ? even **Gemini's**
   median |?start| is 0.98?1.46 s. So tIoU spends most of its budget on a boundary that is *intrinsically*
   undefined to ~1 s. (Owner directive: don't strictly align starts; tolerate ~±2 s.)
2. **Count parity is a deceptive proxy.** mahadevpura-2 hit 1.03× of Gemini's count yet F1@0.5 = 0.150 ?
   right count, wrong boundaries. A count- or overlap-based view hides this.
3. **It flips config rankings on noise.** The hard-cap was golden-neutral at n=6 (+0.006 n.s.) but "best" at
   n=7 once a continuously-active clip entered ? a metric artifact at the 0.5 cliff, not a quality change.
4. **The deficits we must see are start?end-symmetric.** Local's |?start| ? Gemini's; the **entire** gap is
   the rally-**END** (|?end| 2.1?3.1 s vs Gemini ~0.8?1.2 s). A single blended IoU cannot show that ? and
   directing R4 needs exactly that decomposition.

---

2. The metric

2.1 Match rule (asymmetric tolerance ? forgive the start, expose the end)

A predicted window matches a golden rally iff **`|?start| ? ?_start` AND `|?end| ? ?_end`**.
- **`?_start` generous (~2.0 s)** ? absorbs the service-window/annotator start fuzziness (the part that is
  not a model deficit).
- **`?_end` tighter (~1.5 s, but IAA-calibrated ? see §2.4)** ? the rally-end IS a sharp physical event
  (shuttle down/net/out), so a loose end *is* a real deficit we want the metric to expose, not forgive.

> A *symmetric* ±2 s tolerance is tempting (the owner's stated bar) but the preview (§3) shows it is **not**
> automatically more lenient than tIoU 0.5 and it blurs exactly the end-deficit we need to see. Asymmetry is
> the point: relax the dimension that's intrinsically fuzzy (start), keep honest the one that's a real event (end).

2.2 Assignment ? strict 1:1 (Hungarian), never overlap/greedy
Match references to predictions by **global Hungarian assignment** (max total matches under the tolerance
gate), one-to-one. This is what makes **over-production cost precision directly**: `P@? = matched / n_pred`
crashes when the detector emits 75 windows for 52 rallies. (tIoU's overlap matching let extras "match
something"; the W1 over-segmentation pendulum was barely costed.)

2.3 Report the DECOMPOSITION, never one number
1. **`P@?`, `R@?`, `F1@?`** ? precision (over-production), recall (did we find the rally within tolerance), F1.
2. **Boundary-error distributions** ? median **+ IQR** of signed *and* absolute `?start`, `?end` over matched
   pairs, **split**. This is the metric that says *"start solved, end open"* and aims R4.
3. **Over-seg guard (mandatory):** `split_count` (preds covering one ref ? W1 over-seg) and `merge_count`
   (refs under one pred ? the baseline mega-window). The *original* rule was strict per-video: **a promotion
   may not increase either on ANY video.** **REFINED to a NET guard, 2026-06-17 (R1, owner-approved)** ? the
   strict rule is correct for an *incremental* cue but **mis-fires on a structural mega-window?bounded change**
   (splitting a mega-window necessarily lifts `split_count` from ~0, and the raw merge *count* is non-monotonic:
   one window hiding 40 rallies is `count=1, rate=1.0`; bounding it into pieces that each glue 2 neighbours is
   `count=9, rate=0.5` ? FEWER rallies hidden). The net guard scores over-seg on the **normalized rates** as one
   weighted cost per video (`w_merge·merge_rate + w_split·split_rate`, merges weighted higher ? a merge hides a
   rally = recall loss, a split only fragments one), and **passes iff (a) the corpus-mean net cost does not rise
   AND (b) no video's `merge_rate` regresses.** Implemented as `tolerance_metrics.over_seg_guard` (it always
   reports the strict verdict too; revert = read `strict_passes`). See QUALITY_ITERATIONS.md "R1" for the measured
   verdict (milestone: mean `merge_rate` 0.694?0.150, net PASS, strict BLOCK).
4. **?-sweep** (`? ? {1.5, 2.0, 2.5, 3.0}`, symmetric for the trend; asymmetric pair for the headline) ?
   because the convention/IAA noise is large, a single ? misleads (§3). Report the curve.
5. **`tIoU`/`mIoU` demoted to a SECONDARY trend-line diagnostic** (keeps the paper-ablation backbone
   comparable) ? **not the gate.**

2.4 Choosing ? ? it MUST be calibrated to inter-annotator agreement

The preview (§3) shows even **Gemini** scores `tolF1@1.5 = 0.20` on mp2 ? i.e. on ~half the rallies the golden END differs from Gemini's by >1.5 s. That is plausibly **label/convention noise, not model error**: we'd be measuring how two humans (or human-vs-Gemini) disagree on "when did the rally end." **Therefore `?_end` must be ? the inter-annotator end-agreement**, which we do not yet know. **Action:** a small **IAA study** ? a 2nd labeler re-labels 2?3 already-golden videos; set `?_start`/`?_end` to ~the 90th-percentile pairwise human |?start|/|?end|. Until then **freeze provisional ?_start=2.0, ?_end=1.5 and always report the sweep**, with the `? = D·(1??)/(1+?)` derivation recorded so the choice is auditable.

---

3. Preview on the two ground-truth clips (greedy-1:1 approximation; Hungarian in the impl)

Symmetric-? sweep, F1@? (local-best = the per-clip best config: mp2 W1+W2+hardcap, GX010128 W1+W2):

clip / method	tolF1@1.5	@2.0	@2.5	@3.0	(tIoU-F1@0.5)
mahadevpura-2 / **Gemini**	0.203	0.354	0.506	**0.557**	0.557
mahadevpura-2 / local	0.111	0.148	0.222	0.278	0.296
GX010128 / **Gemini**	0.615	0.769	0.813	**0.857**	0.813
GX010128 / local	0.189	0.315	0.346	0.362	0.457

Reading the preview (all honest):
- **`tolF1@~3.0 ? tIoU-F1@0.5`** ? the sweep is a smooth, interpretable knob that *contains* tIoU as a point; good (continuity with the frozen baseline).
- **The local?Gemini gap WIDENS under the tighter, 1:1 tolerance** (GX010128 ratio 0.56 at tIoU0.5 ? **0.41** at ?2.0). R2 charges local for the 1.5× over-production (precision) and the loose ends ? exactly the deficits tIoU under-counted. **This is the metric working, not local regressing.**
- **Even Gemini drops at small ?** (mp2 0.557?0.203 from ?3.0?1.5) ? strong evidence the **golden END convention carries >1.5 s of noise** ? §2.4's IAA calibration is a prerequisite, not a nicety.
- Boundary-error medians (the robust diagnostic, from the validation report): local |?start| 0.78?1.74 s (? Gemini 0.98?1.46) vs local |?end| 2.12?3.12 s (? Gemini 0.76?1.19) ? **the headline R4 signal.**

---

4. Promotion / "trustworthiness" rule (small-n)
A candidate is promotable only if **all** hold:
1. **Paired per-video sign-test sweep on F1@?** is near-unanimous: **6/0 @ n=6 (p=0.031)** or **7/0 @ n=7 (p=0.016)** (8/0 @ n=8, p=0.008); 5/1 or 6/1 is NOT significant. Run on the **growing-golden** set.
2. **No over-seg regression** ? the **net** over-seg guard passes (§2.3.3): the corpus-mean weighted merge/split-*rate* cost does not rise AND no video's `merge_rate` regresses. (Was the strict per-video count rule; refined to net on 2026-06-17 ? see §2.3.3.)
3. **Effect size + bootstrap CI** reported as an honesty band (expect it to span 0 at this n; use it to separate "real" from "directionally-clean-but-negligible," not as the pass/fail).
4. **Stratify, don't blend** ? report the continuously-active blob clip (mahadevpura-1) as its own stratum

  (its near-degenerate scores can swing a small-n mean).
5. **No regression on EITHER eval set** (raw + growing-golden) ? the dual-set discipline.

---

5. Implementation plan (pure, offline, additive ? zero production risk)
- **`backend/eval/tolerance_metrics.py`** (pure, GPU-free, unit-testable):
  - `match_1to1(preds, gts, ?_start, ?_end) -> (matches, P, R, F1)` ? Hungarian via
    `scipy.optimize.linear_sum_assignment` if available, else a pure $O(n^3)$ fallback (corpus n is tiny).
  - `boundary_error_report(preds, gts) -> {?start, ?end: signed/abs median+IQR}` (computed over
    unconditional best-overlap matches internally ? see the refinement note above).
  - `over_seg_report(preds, gts) -> {split_count, merge_count}` (reuse the bipartite logic from
    `segmentation_metrics.merge_split_report`).
- **Wire into `backend/eval/rally_seg_eval.py`** ? emit the R2 block **alongside** tIoU (don't remove tIoU);
  add `--tau-start/--tau-end` (both landed; the ?-sweep helper `tolerance_metrics.tolerance_sweep` exists
  but is not yet surfaced as a `rally_seg_eval` CLI flag). Surface in the experiment leaderboard.
- **`run_regression.py`** ? add the dual-set R2 aggregate + the over-seg-guard gate. *(Status: the guard
  primitive landed as `tolerance_metrics.over_seg_guard` and is wired into `rally_seg_eval.compare()`/`--vs`
  ? the natural home, since `rally_seg_eval` loads the golden manifest + trajectories. The dual-set
  `run_regression.py` aggregate is still a separate future item.)*
- **Tests** ? synthetic intervals: exact-match, start-fuzzy-within-?, end-loose-beyond-?, over-production
  (precision crash), the Hungarian-vs-greedy tie case, empty inputs.
- **Default-additive ? ablation-gated replacement:** R2 is reported next to tIoU (augment). The *gate*
  switches to R2 only after a dedicated **ablation experiment** on the golden corpus shows R2 ranks candidates
  at least as sensibly as tIoU ? so the promotion bar never changes silently mid-flight. When tIoU is retired
  as the gate it is **deprecated with documentation, not deleted** (kept as the secondary trend-line + the
  `QUALITY_ITERATIONS.md` paper-ablation backbone, with a dated note recording the ablation evidence for the
  switch). Owner decision, 2026-06-17 (§7).

---

6. Definition of done
- `tolerance_metrics.py` + tests merged (default-additive); `rally_seg_eval` reports the R2 decomposition
  next to tIoU (the ?-sweep helper exists in `tolerance_metrics` but is not yet wired into the
  `rally_seg_eval` report); the corpus (n?8) re-scored under R2 and recorded in QUALITY_ITERATIONS.
- The IAA study (§2.4) scoped (it can land after R2's code; R2 ships with provisional ? + the sweep).
- `run_regression.py` carries the dual-set R2 gate + the per-video over-seg guard ? but R2 **augments** tIoU
  until the ablation experiment (§7) validates it as the gate; only then does the gate switch.
- tIoU/mIoU retained as a secondary trend-line; when retired as the gate, **deprecated-with-docs (kept for
  posterity)** per the §7 owner decision ? a dated note records why and links the ablation evidence.

---

7. Decisions
Locked (recommended, design-level): Hungarian 1:1 (not greedy/overlap); report the decomposition + sweep,
never one number; tIoU demoted to secondary; R2 default-additive until validated; boundary-error split
start/end is the R4-aiming diagnostic.

Locked (owner review, 2026-06-17):
- **?_end asymmetric (1.5 s) vs ?_start (2.0 s) ? ADOPTED.** Asymmetry is the point: forgive the intrinsically
  fuzzy start, keep the rally-end honest. Your "±2 s, not more" bar maps to `?_start=2.0`; `?_end` stays the
  tighter, IAA-calibrated knob (§2.4).
- **IAA study after R2 code ? ADOPTED.** R2 ships now with provisional `?_start=2.0 / ?_end=1.5` + the full
  sweep; the IAA study lands next to fix ?. R2's code is not blocked on it.
- **Gate: AUGMENT now, then ablation-gate the eventual replacement, then deprecate-with-docs ? ADOPTED.**
  R2 is reported **alongside** tIoU (augment) until validated. tIoU is **not** swapped out silently: before
  R2 becomes the promotion gate, run a dedicated **ablation experiment** on the golden corpus showing R2
  ranks candidates at least as sensibly as tIoU (and exposes the deficits tIoU hid). When tIoU is finally
  retired as the gate, it is **formally deprecated with documentation ? kept for posterity, never deleted**:
  it remains a secondary trend-line and the paper-ablation backbone (`QUALITY_ITERATIONS.md`), with a dated
  note recording *why* it was superseded and the ablation evidence behind the switch. (Consistent with the
  repo's "retain weak/superseded signals; numbers inform, foundations decide" norm.)

10. user (2026-06-21T13:56:29.448Z)

R1 Milestone ? Launch Report (windowing default flip: W1 cap=6 + R4)

Status: LAUNCHED 2026-06-18 (PR #204). **Flip:** the local-windowing defaults go ON ?
`max_window_duration 0?6`, `window_inactivity_end 0?1.5`, `inactivity_rally_thresh 0.08?0.20`,
`inactivity_min_density 0.34?0.30`. **W2 net-crossings, W1.5 hard-cap, and R5 blob-fallback all stay OFF.**
This is the [Launch Report convention](../../) record made **before** the flip merged ? total quality impact
on **both rulers** + the over-seg guard + honest caveats, for posterity.

> **Why this flip is safe to ship:** measured on the **n=13** golden corpus (the grown set, incl. the two
> Boxhill clips GX010137/GX010141), the milestone clears the **net over-seg promotion bar** with room to
> spare, and the verdict was **independently reproduced** (a different harness, #214) and **adversarially
> stress-tested** (5-lens LOO/seed/per-clip workflow). Nothing in the verification refuted it.

---

1. Total quality impact ? BOTH rulers (n=13, leave-one-video-out unit = whole video)

The headline promotion metric is the **R2 task-faithful tolerance-match F1** (`tolF1`, asymmetric ?_start=2/
?_end=1.5); **tIoU-F1@0.5** is kept as the secondary continuity ruler.

config	**R2 tolF1**	**tIoU-F1@0.5**	mean |?end|
baseline (all-OFF)	0.137	0.236	44.6 s
+ W1 cap=12	0.210	0.382	5.7 s
+ cap=6 (R3)	0.320	0.436	5.0 s
+ R4 (milestone, shipped)	**0.359**	**0.474**	**3.3 s**

Paired sign-tests (the promotion evidence):

transition	? tolF1 (95% CI)	sign (W/L/T)	p	? tIoU-F1	sign	p
baseline ? milestone	**+0.222** [+0.152, +0.286]	**12/0/1**	**0.0005**	+0.238 [+0.159, +0.318]	12/0/1	0.0005
cap12 ? cap6 (R3)	+0.110 [+0.070, +0.153]	12/0/1	0.0005	?	?	?
cap6 ? +R4	+0.040 [+0.015, +0.065]	9/1/3	0.0215	?	?	?

The single non-win in baseline?milestone is **mahadevpura_1** (the continuously-active blob: tolF1 0.000?0.000, a tie ? it needs R5, which stays OFF). **Both new clips improve:** GX010137 0.448?0.685, GX010141 0.333?0.431.

2. Over-segmentation guard (the promotion gate)

The R1 **net** over-seg guard (`tolerance_metrics.over_seg_guard`) is the gate (not the strict per-video count rule). Baseline ? milestone:

- **NET verdict: PASS** ? weighted (merge:split = 2:1) merge_rate/split_rate cost **1.303 ? 0.429** (? ?0.874).
- **mean merge_rate 0.651 ? 0.161** ? drops on **all 13** videos; **zero per-video merge_rate regressions**.
- **strict_passes = False BY DESIGN** (not a regression): the raw per-video *count* rule trips because splitting a mega-window lifts split_count from ~0 and the raw merge *count* is non-monotonic (one window hiding 40 rallies = count 1/rate 1.0 ? bounded into pieces gluing 2 neighbours = count 9/rate 0.5 ? **fewer** rallies hidden). The guard scores the recall-meaningful *rate*; see QUALITY_ITERATIONS.md "R1".

3. Verification provenance (independent + adversarial)

- **Independent reproduction (#214 `lovo_resweep.py`, a different harness):** milestone vs baseline 0.137?0.359, sign **12-0-1, p<0.001**; R4 on-vs-off 0.320?0.359, sign **9-1-3, p=0.021** ? matches to 3 decimals. Its LOVO cap-selection picks **cap=6 unanimously (13/13)** on tIoU.
- **5-lens adversarial workflow** (reproduce / leave-one-out / R4-loss audit / new-clip sanity / guard audit ? synthesis): verdict **STANDS**. Every LOO fold (n=12) clears CI-excludes-0 AND p<0.05; worst fold (drop GX010128, the biggest gainer) still ?+0.202, p=0.0010 ? **no single clip drives the win**.

4. Honest caveats (recorded so they aren't relitigated)

1. **The W1 cap is the dominant lever; R4 is a marginal increment.** baseline?cap6 is ~+0.18 of the +0.222 tolF1 gain; the R4 inactivity-end trim adds a *small but significant* +0.040 (p=0.0215, sign 9/1/3). Frame R4 as "a small, significant end-precision gain" (mean |?end| 4.96?3.31 s), not a co-equal driver.
2. **R4's one loss is GX010137** (tolF1 0.727?0.685, ?0.042) ? a tolerance-accounting artifact, **not** an R4

failure: on that clip strict F1 *improves* (0.805?0.822), |?end| *improves* (0.447?0.439), over-seg unchanged;
R4 simply trimmed 4 trailing predicted segments (43?39, toward gt=34). The clip's *overall* milestone delta is
a large win (+0.237).
3. **GX010141 is the worst-behaved clip post-milestone** (merge_rate still 0.448, ~30 s windows vs gt max ~10.5 s)
? partially-merged, **not** a regression (it improves +0.098) and it does not break the guard, but it's a
future over-merge target as the corpus grows.
4. **cap=4-vs-6 wobble on tolF1** (from #214): cap=4 edges cap=6 by ~0.005 (noise) ? but **LOVO re-fitting to
cap=4 generalizes *worse*** (??0.004) and tIoU is firmly cap=6, so **cap=6 stands**. Watch as n grows; switch
only if cap=4 pulls ahead *consistently across folds*.
5. **n=13 is small.** "Robust" here means "no single clip or bootstrap seed flips the verdict," not large-sample
certainty. (Bootstrap CIs jitter ~0.001?0.004 run-to-run; means/sign-tests/p-values are exact.)
6. **Nightly re-baselined + an all-OFF floor added (DONE).** The flip made the nightly's `default` config
(= SERVED) equal `milestone`, so its config-drift detector flagged the frozen baseline STALE ? re-snapshotted
in #216. A pinned **`baseline_off`** reference config was then added (follow-up) so every nightly report keeps
showing the delta the milestone buys (floor tolF1 **0.137** vs milestone **0.359**) ? the launched win has to
keep standing its ground, not be taken at launch-time value.
7. **Pure-logic eval only** (CPU, persisted trajectories) ? no GPU/real-video re-detection was run; the flip
changes only the windowing knobs the served path already reads.

5. What did NOT flip (still default-OFF, retained)

- **W2 net-crossings gate** ? no ground-truth-backed lift (the weak-signals-retained directive keeps it).
- **W1.5 hard-cap** ? over-segments normal clips (chops before R4); superseded by R5 for the blob case.
- **R5 blob-fallback** ? a stratum tool for continuously-active clips; turn on selectively, not globally.

11. assistant (2026-06-21T13:56:32.825Z)

Now I have a clear picture of where R-series stands (R1 shipped, R2 implemented as augmenting). Let me quickly read the remaining track docs to confirm Gate-A wiring status, harness, golden fixtures, and the QUALITY_ITERATIONS for the latest measured state, plus check the §7 tracker freshness against the docs.

12. user (2026-06-21T13:56:33.822Z)

Multi-signal rally fusion ? add a human (person) cue alongside the shuttle

Status: **P1 DONE** (2026-06-13) ? the person cue is extracted to
`backend/pipeline/detectors/person_detector.py` (`PersonDetector`), consumed by `yolo_hybrid` with no
behavior change. **P2?P4 remain owner-gated.** Phases below are sequenced but each is a separate PR.
Owner decisions captured in [§9](#9-open-decisions-for-the-owner).

> **Companion docs:** the 2-layer mental model is HOW_RALLY_DETECTION_WORKS.md;
> the modular-fusion architecture this builds on is **ADR-007** (archives/decisions/DECISIONS.md);
> the permissive-dependency rule is **ADR-002 / ADR-005**; the audio cue (the #1 next

cue, already
> scaffolded) is archives/research/AUDIO_RALLY_DETECTION_PLAN.md.
> The vendor labels that gate the learned path come from [GOLDEN_SET_VENDORING.md](archives/roora-vendor/GOLDEN_SET_VENDORING.md)
> and the labeling spec in [ROORA_LABELING_GUIDELINES.md](archives/roora-vendor/ROORA_LABELING_GUIDELINES.md).
> **How each cue here is added/tested without production regression** ? A/B, shadow eval, and the
> promote-only-on-lift discipline ? is EXPERIMENT_HARNESS.md (every P1?P4 cue
> below lands flag-gated, default OFF, promoted only on measured LOVO lift).

---

1. Context ? why this, why now

Rally detection today runs **one cue at a time**. Each segmenter independently turns a single signal
into rally windows:

Segmenter	Signal	File
`wasb_hybrid` / `tracknet_hybrid`	shuttle velocity (frozen WASB/TrackNet trajectory)	`backend/pipeline/detectors/tracknet_runner.py`
`yolo_hybrid` (the Pydantic `default_segmenter` fallback; `config.json` overrides it to `gemini`)	person velocity (torchvision + ByteTrack)	`backend/pipeline/segmenters/yolo_hybrid.py`
`motion`	raw pixel motion	`backend/pipeline/segmenters/motion.py`
`gemini`	a cloud VLM watching the whole clip	`backend/providers/gemini.py`

There is no fusion. The owner's directive: make the detector **use the shuttle *and* the humans
together** ? and be ready to fold in more signals (audio, later pose) ? delivered
MIT / Apache-2.0 / BSD-clean so we can ramp commercially without a licensing rethink.

This is **not greenfield** ? it operationalizes work the repo already blessed:

- **`HOW_RALLY_DETECTION_WORKS.md` open question:** *"Fusion: combine shuttle-motion (WASB) with
  player-motion for static cams."*
- **ADR-007** already designed a **modular "rally layer"** that fuses cues *outside* the frozen WASB
  model, with **audio as the #1 next cue and person/pose parked as #2**. This plan picks up the parked
  person cue and generalizes the fusion seam so future cues are "register a producer," not "re-touch
  the segmenter."
- **`backend/pipeline/detectors/rally_gate.py` already exists** ? the net-crossing gate, explicitly
  labelled in its own docstring as *"Gate A in the over-fire fix spectrum (**B = player-stance state
  machine, future**)."* The person cue in this plan **is that Gate B ? now built as the `FusionSegmenter` person path (default-OFF + unmeasured on golden; see §10 P2).**

The owner's "held-shuttle" false positive ? what's actually going on

The owner observed a "rally" that was just a player **holding / flicking the shuttle in hand** between
points. This is the *exact* failure `rally_gate.py` was written to kill:

> *"WASB over-detects: when a player flicks/tosses the shuttle back to the opponent for service after
> losing a point, that single trajectory looks like a rally. The discriminator is

structural: a real
> rally has the shuttle crossing the net MULTIPLE times; a single service feed crosses ~once."*

The structural fix already exists and is validated offline ? but it is NOT wired into the production runtime path. `rally_gate.apply_gate(..., min_crossings=2)` is called only from the **eval drivers** (`eval_partial_labels.py`, `export_wasb_segments.py`, `calibrate_local.py`; `distill_local.py` uses `rally_gate.gate_windows` to compute the `net_crossings` feature, not `apply_gate`); the live segmenters historically didn't apply it. P2 closed this: `FusionConfig.min_crossings` now exists (default `0` = OFF), so the served Gate A is a config flag ? consistent with §10 P2 below, marked DONE. So the single cheapest, highest-confidence win in this whole plan is **P2's first step: fold the existing Gate A into production.** Everything else (the person cue / Gate B, the learned fusion) is robustness on top of that.

Why humans help ? and the honest risk

The beachhead videos are **static tripod** (Bangalore academies, [PMF_MARKET_RESEARCH.md](archives/field-pmf/PMF_MARKET_RESEARCH.md)), so player presence/motion is a usable cue. ADR-001's warning was specifically about **broadcast pan** cameras, where player-motion fooled the old heuristic ? *that failure mode does not apply to a fixed court cam.* But we keep **the shuttle as the camera-invariant primary** so the general bet stays robust on other camera types. Humans then add:

- **Precision** ? reject shuttle-on-floor / warm-up knock-ups / held-shuttle feeds when the court
  isn't occupied by ?2 players in a play posture.
- **Recall** (via the learned classifier) ? features that help bridge shuttle occlusion/blur gaps.

---

2. Core architecture ? one rally-fusion layer, cues are swappable producers

Formalize ADR-007's diagram. Every cue is a **black box that emits a time-aligned signal** (per-frame activity and/or per-window features); **fusion happens in *our* layer, never inside a frozen model.**

```
frames ?? WASB (frozen, MIT)      ?? shuttle trajectory ???
frames ?? person detector (ours)  ?? tracks + boxes ???????
audio  ?? hit detector (ours)     ?? hit events ????????????? RALLY LAYER (ours) ?? rallies
[frames ?? pose/stance (ours; parked #3) ?? stance ????????   feature + decision fusion
```

- **Shuttle = primary, camera-invariant.** Windows still **seed** from
  `TrackNetRunner.trajectory_to_action_windows()` (`tracknet_runner.py:234`). Other cues *enrich and
  adjudicate* shuttle-seeded windows; they do **not** replace the shuttle seed. Person runs only over
  shuttle-seeded candidate windows + padding, not the whole video ? which bounds the added cost.
- **A small `RallyCue` contract.** Each producer yields `(frame_idx ? features)` + per-

window stats.
Adding a cue = register a producer; the existing `SegmenterRegistry` (`segmenters/base.py:35`) is the template for the pattern. Keep the single-cue segmenters registered as fallbacks.

---

3. The two fusion modes (both ? per the owner decision)

3.1 Decision-level gate (safe baseline, ships first, no labels needed)

```
rally_active = shuttle_motion ? (?2 persons in court zone ? recent person track)
```

The `? recent-track` clause tolerates 1?2 frame person-detector misses so the AND-gate can't cause a recall cliff. This composes with the **existing net-crossing Gate A**:

```
keep window  ?  net_crossings ? 2            (Gate A ? EXISTS in rally_gate.py, eval-only today)
                ?  (?2 in-court players ? recent track)   (Gate B ? the person cue, BUILT default-OFF, unmeasured on golden)
```

Gate A kills the service-feed / held-shuttle case structurally; Gate B adds court-occupancy evidence for the cases a single trajectory can still fool (e.g. a knock-up that does cross the net). No training data required ? this is the label-free fusion that ships before any golden labels exist.

3.2 Feature-level (primary, the learned path)

The cues emit a **small set of aggregated, scale/translation-invariant per-window features ? NOT raw coordinates.** Raw pixel coordinates would memorize *this* court/camera and fail to generalize off a still-small labeled set; counts / ratios / zone-membership / two-sidedness transfer.

What already feeds the classifier (`distill_local.py:40`, `FEATURE_NAMES`): `duration, peak_velocity, mean_velocity, active_ratio, net_crossings` ? note **`net_crossings` is already a feature** (computed via `rally_gate.gate_windows`).

Person features to add (~4, deliberately few):

Feature	Meaning	Rally signal
`players_in_court`	median count of persistent in-court tracks	? 2 (singles) / 4 (doubles); 0?1 = dead time
`two_sided_motion`	are moving players on **both** net-halves?	rally = two-sided; one person feeding = one-sided
`mean_player_motion`	aggregate normalized player velocity	rally = sustained; resting/standing = low
`in_court_ratio`	fraction of frames with ?2 in-court players	track stability vs flicker
`shuttle_player_colocation`	fraction of frames the shuttle sits inside/adjacent to a person box (hand region)	**held = high; in-flight = mostly free space** ? the direct held-shuttle discriminator

These five person features are now **built** (`fusion_features.py::PERSON_FEATURE_NAMES`; default-OFF, unmeasured on golden) and join the shuttle features in the candidate `stats` JSON (`database.py::add_candidate_segment`) and the feature vector of the **existing distilled classifier ? the tiny numpy-only L2 logistic regression in `backend/eval/classifier.py` (ADR-004), NOT a new net.** It scores each candidate window `P(real rally)` and replaces the Gemini call at runtime.

What the model learns: the **weights** on those few features ? e.g. down-weight a shuttle-motion window with 0?1 in-court players / one-sided motion / high shuttle-colocation (warm-up, floor shuttle, held-shuttle feed); up-weight ?2 players, two-sided, sustained motion, shuttle in free flight.

Small-N discipline (non-negotiable):
- Keep to **~4 person features** ? more features on a still-small labeled set overfit; that's exactly why the model stays a logistic regression, not a net.
- Use the **leave-one-video-out (LOVO)** protocol that `distill_local.py` **already implements**: train on the other videos' candidates, predict the held-out *video*, score vs its labels ? never a held-out *window* from the same video (windows in one video are correlated; that flatters the score).
- Training labels today are **Gemini silver labels** (distilled offline, ADR-004/007 ? Gemini is a teacher, never a runtime dependency, never a training-data *source* for owned weights); **golden labels are the honest measuring stick**, and as the Roora golden corpus grows they can also become training labels. Until the labeled set is larger, treat the learned weights as **calibration / direction-check, not "a fully general model"** ? the decision-level gate (§3.1) is the shippable, label-free fusion in the meantime.

> *(Gated on golden labels existing ? the same blocker ADR-007 flags for audio. See
> [GOLDEN_SET_VENDORING.md](archives/roora-vendor/GOLDEN_SET_VENDORING.md).)*

---

4. Shuttle-state: in-flight vs in-hand (the misclassification, mapped to real code)

The "shuttle moved" seed conflates a shuttle **in flight** with a shuttle **carried in hand**. The discriminators, and where each lives:

Signal	In-flight	Held / static	Status
`net_crossings`	? 1 (usually several)	~0 (held) / ~1 (single feed)	**EXISTS** (`rally_gate.py`); served Gate A wired via `FusionConfig.min_crossings` (default 0 = OFF)
`shuttle_player_colocation`	low (free space)	high (near a hand)	**BUILT** (`fusion_features.py`; default-OFF under the person cue, unmeasured on golden)
`direction_reversals`	several (racket contacts)	~0	derivable from trajectory; partial overlap with crossings
`peak shuttle speed`	high	walking speed	derivable from existing velocity

So the in-flight-vs-held fix is **mostly Gate A (already built) + the new colocation feature from the person cue**. **No new label column is needed** ? the golden labels already start a rally at the *serve-contact frame* and treat all pre-serve / held-shuttle time as dead time (see [ROORA_LABELING_GUIDELINES.md](archives/roora-vendor/ROORA_LABELING_GUIDELINES.md)), so held-shuttle moments are negatives by construction and are the ground truth that proves the detector wrong.

---

5. Adversarial mitigations (integrating safely)

- **Court-region mask** ? polygon around the court; drop persons outside it (spectators, coaches, adjacent courts). Config-supplied per camera; ~1 ms. Source: a manual per-camera polygon first; the **4 court corners + net-line** that Roora labels once per video (see the labeling doc) build this mask **and** the net split that `two_sided_motion` / `net_crossings` need.
- **Player-count constraint** ? expect **2 (singles) / 4 (doubles)** persistent in-court tracks; suppress transient/extra detections. The `singles_or_doubles` manifest flag sets the expectation.
- **Temporal association** ? reuse `supervision` **ByteTrack** (already in `yolo_hybrid`); reject singleton detections; smooth across frames.
- **Multi-court rejection** ? proximity to the primary court region (homography optional, later).
- **Domain shift (COCO frontal ? side/overhead court)** ? start **box-only** (robust); keep the detector swappable. Fine-tuning a person detector is **parked** ? we have no in-house person labels, and the **paid-LLM/training guardrail + minors-exclusion** apply to any future person-training data.
- **Fusion-degradation guard** ? shuttle stays primary; person is gate + feature, never the sole trigger; the single-cue segmenters remain registered fallbacks.

---

6. Box vs pose

Box-first. Occupancy + count + two-sided motion + shuttle-colocation is enough for both the gate and the features, and box detection is the cheap, robust win. **2D pose is parked behind the same `RallyCue` interface** ? pose (MediaPipe Pose / RTMPose, both Apache-2.0) buys serve-ready stance gating and shot cues later, at the cost of a second model. This matches ADR-007 parking pose as the #2/#3 cue.

---

7. Performance

The person detector is the **existing in-process torchvision model** run at **frame-skip 3?5** (`yolo_frame_skip` default 3) with track interpolation between sampled frames. The shuttle keeps running in its separate WSL process ? the two are **separate processes**, so a "shared backbone" isn't free; budget realistically as the existing `yolo_hybrid` cost minus skip savings. Person runs **only over shuttle-seeded windows + `ai_handoff_padding`**, not the whole video, which bounds the cost.

---

8. Licensing (the guardrail table ? all green)

Every dependency stays **MIT / Apache-2.0 / BSD** (ADR-002 / ADR-005; `ultralytics` was dropped for AGPL-3.0).

Cue / model	Code	Weights / data	Verdict
Shuttle: WASB-SBDT, TrackNetV3	MIT	frozen badminton weights	? (ADR-001/002)
Person (default): torchvision Faster R-CNN / RetinaNet / FCOS	BSD/MIT	COCO-pretrained	? already in repo (`_DETECTOR_FACTORIES`)
Person (escalation): **YOLOX**, **RT-DETR / RT-DETRv2**	Apache-2.0	COCO / Objects365	?
Pose (parked): **MediaPipe Pose**, **RTMPose / MMPose**	Apache-2.0	COCO-structure	?
Tracking: `supervision` ByteTrack	MIT	n/a	? already in repo
Banned	**Ultralytics YOLOv5/8/11 (AGPL-3.0)** · **MonoTrack (Adobe, non-comm)** · **YOLO-NAS (Deci, non-comm)**	?	?

COCO caveat to record: COCO *annotations* are **CC BY 4.0** (commercial OK *with attribution*);
COCO *images* are Flickr-ToS (per-image). Pretrained-**weight** use is clean; record the attribution.

---

9. Open decisions (for the owner)

1. **Court-region mask source** ? manual per-camera polygon (simple, ship first) vs auto court
   homography/line detection (later). **Recommend manual first**, fed by Roora's court-corner labels.
2. **Person detector default** ? keep torchvision (zero new deps, BSD/MIT) vs adopt YOLOX/RT-DETR
   (Apache, faster) now. **Recommend torchvision first**, escalate only if eval shows a speed/accuracy gap.
3. **Pose now or parked** ? **recommend parked** behind the interface (box-first).
4. **Golden labels** ? feature-level fusion (P3) is **blocked on golden rally labels**; the source is
   the VLC `rally-annotator` + the Roora vendor pilot ([GOLDEN_SET_VENDORING.md](archives/roora-vendor/GOLDEN_SET_VENDORING.md)).

---

10. Phased execution (owner-gated; ONE PR per slice, off `master`)

- **P0 ? this design doc** + [ROORA_LABELING_GUIDELINES.md](archives/roora-vendor/ROORA_LABELING_GUIDELINES.md), indexed in
  `docs/README.md`, cross-linked from `HOW_RALLY_DETECTION_WORKS.md` and ADR-007. *(this deliverable;
  docs-only, no code.)*
- **P1 ? extract the shared person cue:** ? **DONE (2026-06-13).** Lifted the torchvision detector +
  ByteTrack + velocity out of `yolo_hybrid` into `backend/pipeline/detectors/person_detector.py`
  (`PersonDetector`); `yolo_hybrid` now orchestrates only and delegates to `self.person` (no behavior
  change ? regression-safe, proven by the unchanged process_video output tests). New
  `tests/test_person_detector.py`; suite +1 green; person_detector is type-clean (left mypy quarantine
  for the moved code). The fusion segmenter (P2) reuses this same producer over shuttle-seeded windows.
- **P2 ? the FusionSegmenter (Gate A + Gate B + person-feature persistence):** ? **DONE (2026-06-14).**
  Shipped `backend/pipeline/segmenters/fusion_hybrid.py::FusionSegmenter` (registry `fusion_hybrid`,
  default-OFF ? `default_segmenter` unchanged), subclassing the trajectory engine via two identity
  hooks (`_enrich_candidate_stats`, `_select_for_handoff` ? wasb/tracknet stay byte-identical, adversarially
  verified). It seeds windows from the shuttle substrate, **always persists

`net_crossings`** (Gate A signal),
  and ? when `indexing.fusion.cue_flags.person` is on ? runs the person cue over each window's frames
  (`PersonDetector.detections_per_frame`) and persists the ~5 person features (`fusion_features.py`,
  pure + unit-tested). Served gating is optional + default-OFF: `fusion.min_crossings` (Gate A) /
  `fusion.min_players` (Gate B) skip sub-threshold windows from the handoff while **persisting the full
  superset** (offline substrate intact). Court mask = optional `fusion.court_polygon` (None ? count-only).
  Lift to be measured via the harness; held-shuttle-drop + person-feature values are GPU/owner-side.
- **P3 ? feature-level (learned) fusion:** add the person features (incl. `shuttle_player_colocation`)
  to `distill_local` / `classifier`; recalibrate IoU on the golden set; report **LOVO lift on held-out
  videos**. *(Gated on golden labels.)*
- **P4 ? additional cues:** wire **audio** (ADR-007's #1, already scaffolded in
  `backend/pipeline/audio/hit_detector.py`) and later **pose**, through the same `RallyCue` interface.
  Each is add-a-producer, not a refactor.

---

11. Verification

- **Offline, runnable in this dev container (no GPU/torch/cv2):** unit tests for the fusion gate logic
  and `RallyCue` aggregation (court-mask filter, player-count constraint, `? recent-track` tolerance)
  and the shuttle-state signals (net-crossing count ? extends the existing `rally_gate` tests ? and
  person-colocation) with **synthetic trajectories + person boxes**. A key-license assertion test
  (no `ultralytics` import; detector factories stay torchvision/Apache).
- **Machine-side (owner, GPU/WSL ? the full pipeline can't run in this container):** run the
  `FusionSegmenter` on a real static-tripod clip; confirm **held-shuttle / warm-up / spectator false
  positives drop** vs shuttle-only (this should reproduce the owner's observed bug and show it fixed ?
  start by confirming Gate-A-in-production alone fixes it); **LOVO IoU lift on the golden set** for the
  learned path.
- **Docs (P0):** links resolve; both docs indexed in `docs/README.md`; this doc cross-linked from
  `HOW_RALLY_DETECTION_WORKS.md` (the fusion open-question) and ADR-007.

13. user (2026-06-21T13:56:34.259Z)

How rally detection works (the 2-layer mental model)

A short conceptual map of how we go from raw video to rally `[start, end]` segments ?
and, crucially, **which layer our annotation can improve.**

```
  video ??? frames ??? [DETECTION] ??? shuttle (x,y,visible) per frame
                              ?                     ?
                         (WASB CNN,          per-frame trajectory
                          frozen weights)          ?
                                                   ?
                                          velocity signal  (how fast the shuttle moves)
                                                   ?
```

```
                                        [SEGMENTATION]  ??? the 3 knobs we tune
                                                ?
                                                ?
                                   rally windows  [start, end] ??? IoU vs golden labels
```

Two layers ? keep them separate in your head

1. DETECTION ? "where is the shuttle, this frame?"
WASB is a CNN that, for each frame, outputs a heatmap ? the shuttle's `(x, y)` + a visibility/confidence. Its weights were learned (by gradient descent, on shuttle-coordinate labels) and we use them **frozen**. We chose a *shuttle* detector on purpose: "shuttle in flight" is the signal for "a rally is happening," and it's **camera-agnostic** ? unlike player-motion, which fooled the old heuristic on broadcast cameras that pan with the players. (Your videos are static tripod, so player-motion would also work ? but shuttle-motion is the robust general bet.)

2. SEGMENTATION ? "when is a rally?"
From the per-frame `(x, y)` we compute the shuttle's **velocity** (displacement between frames). Then three knobs turn that into rally intervals:
- **`velocity_thresh`** ? a frame is "active" only if the shuttle is moving faster than this (in flight, not lying on the floor / being picked up).
- **`merge_gap`** ? active frames within this many seconds get joined into one rally (bridges brief gaps: a shuttle momentarily occluded, or the beat between two hits).
- **`min_window_duration`** ? throw away windows shorter than this (blips / false flickers).

Output: a list of `[start, end]` windows = the rallies.

Where the annotation fits (the key point)
Our Roora/golden labels are **rally `[start, end]` times** ? they say *when* rallies are, not *where the shuttle is*. So they can't teach WASB's detector (that needs `(x,y)` labels). They **can** grade and tune the **segmentation layer**: we score predicted windows against the golden rallies with **temporal IoU** (how much they overlap), and pick the 3 knobs that maximise it ? **calibrate on some rallies, measure on held-out ones** so the number is honest.

Where sparks could land (open questions ? not commitments)
Things worth a shower-thought; any of these could beat 3 hand-tuned thresholds:
- **Richer features than raw velocity:** acceleration, trajectory *continuity* (a real rally is a smooth arc that survives occlusion), shuttle *height/arc*, or **direction reversals** (each reversal ? a shot ? ties to `shots_count`!).
- **A tiny learned segmenter** instead of 3 thresholds: feed the trajectory features to a small model trained on the golden rally labels (still the segmentation layer, just smarter).
- **Hysteresis** (two thresholds: enter-rally high, stay-in-rally low) ? robust to mid-rally dips.
- **Rhythm/periodicity:** rallies have a hit cadence; dead time doesn't.
- **Fusion:** combine shuttle-motion (WASB) with player-motion for static cams. ? designed in MULTI_SIGNAL_FUSION_PLAN.md (one rally-fusion layer; the person cue is

ADR-007's parked #2 and `rally_gate.py`'s "Gate B" ? now built (default-OFF, unmeasured on golden) in `fusion_hybrid.py`).
- **Per-video calibration as a *feature*:** "watch a few labeled rallies ? auto-tune to this
court/lighting ? segment the rest sharply" ? exactly your earlier idea, at the right layer.

The whole point of per-video calibration: a handful of *your* labeled rallies nudges these knobs
to *your* court/camera, and we report the IoU lift on the rallies it never saw.

14. assistant (2026-06-21T13:56:37.907Z)

Now let me read the remaining docs: EXPERIMENT_HARNESS, GOLDEN_REGRESSION_FIXTURES, ROORA_LABELING_GUIDELINES, and check QUALITY_ITERATIONS for the latest measured state to validate the §7 tracker freshness.

15. user (2026-06-21T13:56:38.877Z)

Experiment harness ? A/B + shadow eval for rally-detection quality (no-regression by design)

Status: **P1 IMPLEMENTED** (`backend/eval/experiment.py` + `tests/test_experiment.py`, shipped
2026-06-13) ? the `Variant` layer, `compare()` (labeled A/B, LOVO-honest), `compare_unlabeled()`
(SxS on NEW unlabeled footage), and the leaderboard. **P2?P4 remain owner-gated.** Phases below are
sequenced but each is a separate PR.

> **Companion docs:** the cues this experiments *on* are in
> MULTI_SIGNAL_FUSION_PLAN.md; the scoring mechanics are
> EVALUATION.md; the no-regression gate + golden labels + `baseline.json` come from
> [GOLDEN_SET_VENDORING.md](archives/roora-vendor/GOLDEN_SET_VENDORING.md); the modular-cue architecture is **ADR-007**
> (archives/decisions/DECISIONS.md); the distilled classifier is
> **ADR-004**. The 2-layer mental model is HOW_RALLY_DETECTION_WORKS.md.

---

1. Context ? why this, why now

PR #112 (merged) added the multi-signal fusion design ? shuttle + person + audio cues, all **default
OFF**. The owner's directive for *how* we evolve quality from here:

> *"Evolve the codebase so we can keep finding new ways to improve quality and experiment with them ?
> or add them as a signal ? without suffering any regressions if the idea doesn't work out. Rolling
> back the code would be too dangerous."*

So **experimentation and deployment must be decoupled.** The contract:

1. Merge experimental code freely ? it's **gated OFF**, so it cannot change production behavior.
2. Test it **offline against golden labels** (most experiments never touch the served path at all).
3. Promote a variant to default **only on measured lift**, through a CI eval gate.
4. **"Rollback" = flip a config flag, never `git revert`.** The proven path is a pinned variant that is

always present.

The pleasant surprise (Explore audit, 2026-06-13): ~80% of this harness already exists and is simply not composed *as a system*. The missing piece is a thin **variant/experiment layer + the discipline**, not a platform. §4 is the exact code surface so the next session does not re-discover it.

---

2. The thesis ? separate expensive DETECTION from cheap DECISION

The expensive, GPU/WSL-bound, risky part is **detection**: "what signals exist per frame" (shuttle WASB trajectory, person tracks/boxes, audio hits). The cheap, swappable, pure-Python part is the **decision**: "given those signals, where are the rallies" (windowing + gates + the classifier).

> **Run detection once per video, persist all per-cue signals + per-candidate features, then re-score
> any number of variants OFFLINE ? deterministically, with no GPU and zero production risk.**

This is not aspirational ? `distill_local.py` and `calibrate_local.py` already do exactly this on stored WASB trajectories. The candidate-segments table (`stats` JSON) is the persistence substrate.
? Most experiments are pure re-scoring of stored data; they never enter the served pipeline.

---

3. What already exists (the audit ? do NOT rebuild)

Exact public surface, with `file:line`. **This table is the anti-duplicate-work artifact** ? start here.

3.1 Scoring (variant-agnostic ? grades any `List[Interval]`)
`backend/eval/metrics.py`
- `interval_iou(a, b) -> float` (:15) ? temporal IoU of two `(start,end)` intervals.
- `match_intervals(preds, gts, iou_threshold=0.5) -> MatchResult` (:48) ? greedy 1-to-1 IoU matching.
- `score(preds, gts, iou_threshold=0.5, match_result=None) -> Score` (:104) ? P/R/F1, mean IoU, boundary errors; `Score.as_dict()`.
- `pr_curve(preds, gts, thresholds=None) -> List[Score]` (:138).

3.2 A/B + cross-validation primitives (the comparison engine already exists)
`backend/eval/calibration.py`
- `improvement_report(predict_fn, baseline_config, tuned_config, gts, iou_threshold=0.5) -> dict` (:148) ? **before/after metrics + %?. This IS the A/B delta.**
- `leave_one_video_out(videos, configs, metric="f1", iou_threshold=0.5) -> dict` (:113) ? **LOVO: pick on other videos, score on held-out video (honest cross-video).**
- `cross_validate(predict_fn, configs, gts, k=5, metric="recall", iou_threshold=0.5) -> dict` (:72) ? within-video k-fold overfit guard.
- `select_best(predict_fn, configs, gts, metric="f1", iou_threshold=0.5) -> (Config, float)` (:61).
- `evaluate(preds, gts, iou_threshold=0.5) -> Score` (:41); `kfold_indices(n, k)` (:46).
- LOVO video dict shape: `{"name": str, "predict_fn": PredictFn, "gts": [Interval]}`.

3.3 No-regression gate + baselines (the promotion gate already exists)
`backend/eval/regression.py`
- `regress(db, video_id, ground_truth, stage="final", iou_threshold=0.5, detector=None, tolerance=DEFAULT_TOLERANCE, baseline_path=DEFAULT_BASELINE_PATH) -> RegressionResult` (:104) ?

score stored preds vs GT, flag if P **or** R drops > tolerance.
78 - `regress_with_provider(...)` (:160) ? same, fetching GT via the annotation provider.
79 - `save_baseline(video_id, stage, precision, recall, f1, ..., gt_fingerprint=None)` (:80) · `load_baselines(path) -> dict` (:68).
80 - `RegressionResult` (:43): `regressed`, `reasons`, `gt_hash`, baseline P/R, `tolerance`, `has_baseline`; `as_dict()`.
81 - Default baseline file: `eval_baselines/regression_baseline.json` (:33).
82
83 `run_regression.py` ? CI-gatable CLI, `main(argv)` (:52). **Exit codes: 0 ok / 1 regression / 2 no-DB / 3 no-baseline.** Flags: `--video-id --tier --annotations --stage{candidates,final} --detector --iou --tolerance --baseline --update-baseline --allow-missing-baseline --json`.
84
85 ### 3.4 Pinnable model artifact
86 `backend/eval/classifier.py` ? `LogisticRegression` (pure numpy, L2): `fit(X,y,feature_names=,class_weight="balanced")` (:40), `predict_proba` (:75), `predict(threshold=0.5)` (:81), `to_dict`/`from_dict` (:86/:97), `save`/`load` (JSON) (:107/:111). **A trained model = one JSON file ? a pinnable, hashable variant artifact.**
87
88 ### 3.5 Offline re-scoring substrate (persist once, re-score forever)
89 `backend/infrastructure/database.py`
90 - `add_candidate_segment(video_id, detector, start_time, end_time, chunk_index=None, confidence=None, stats=None, detection_params=None) -> Optional[int]` (:318) ? `stats`/`detection_params` stored as JSON; `candidate_segments` table cols incl. `stats`, `detection_params`, `human_label`, `confirmed_segment_id` (schema :97).
91 - `query_candidate_segments(video_id, detector=None, only_unlabeled=False) -> List[dict]` (:355) ? returns rows with deserialized `stats`.
92
93 ### 3.6 Label assignment + feature glue
94 `backend/eval/training.py`
95 - `label_candidates(candidate_intervals, gt_intervals, iou_threshold=0.5) -> List[int]` (:50) ? y-labels by IoU match to golden (same matcher as the evaluator).
96 - `build_training_set(candidate_rows, gt_intervals, iou_threshold=0.5, feature_fn=extract_yolo_features) -> (X, y, intervals)` (:64) ? pluggable `feature_fn`.
97
98 ### 3.7 Existing LOVO drivers to mirror (not duplicate)
99 - `backend/eval/distill_local.py` ? `run(specs, iou=0.5, threshold=0.5, model_out=None)` (:98), CLI `python -m backend.eval.distill_local --manifest videos.json --model-out ...`; `FEATURE_NAMES = [duration, peak_velocity, mean_velocity, active_ratio, net_crossings]` (:40); already does LOVO over Gemini silver labels.
100 - `backend/eval/calibrate_local.py` ? `run(specs, iou=0.5, metric="f1")` (:77); `LocalConfig = (velocity_thresh, merge_gap, min_window_duration, min_crossings)`; grid + LOVO.
101 - `backend/eval/calibrate_wasb.py` ? `run(...)`, `load_golden_gts()`; `backend/eval/gt_loader.py::load_gt_intervals(csv_path, *, strict=True)` is THE canonical golden reader.
102
103 ### 3.8 Registries + config knobs (variant selection)
104 - `backend/pipeline/segmenters/base.py` ? `SegmenterRegistry` (:35), singleton `segmenter_registry` (:63); registered (6): `motion`, `gemini`, `yolo_hybrid`, `tracknet_hybrid`, `wasb_hybrid`, `fusion_hybrid`.
105 - `backend/annotations/base.py` ? `AnnotationProviderRegistry`, singleton `annotation_registry`; providers `file`, `auto`.
106 - `backend/config/models.py::IndexingConfig` ? `default_segmenter` (:65, default `"yolo_hybrid"`), `detector_model` (:72), `detector_score_threshold` (:73), `yolo_velocity_threshold` (:74), `yolo_frame_skip` (:75), `min_segment_duration` (:68), `dead_time_merge_gap` (:69), `motion_threshold` (:67). The `min_crossings` knob now lives on `FusionConfig` (default `0` = OFF; also `min_players`, both served Gate A/B thresholds ? see fusion P2).
107
108 ### 3.9 Confirmed ABSENT (no duplication risk)
109 Explore found **no** existing experiment / variant / shadow / A-B *machinery* ? only a stray "A/B test"
110 aspiration comment in `backend/pipeline/detectors/base.py` and an "experiment E1" note in
111 `archives/research/AUDIO_RALLY_DETECTION_PLAN.md`. The `regress()` + baseline path is

the canonical comparison mechanism.

---

4. The variant abstraction (the one thin thing to add)

A **`Variant` = a declarative recipe**, not a code branch:

```
Variant = (segmenter_name, IndexingConfig overrides, enabled cues + per-cue params, classifier model path/hash)
```

JSON-serializable; selected via the existing `segmenter_registry` + `IndexingConfig.default_segmenter`.
Control = a *pinned, frozen* variant (recipe + model hash) that is always registered and is the
default. Adding a new cue = a new Variant differing by one flag ? the control recipe is untouched.

---

5. Three experiment modes

5.1 Offline A/B (primary, label-driven, zero production risk)
`compare(videos, variant_A, variant_B)` ? a thin driver composing **existing** functions:
`query_candidate_segments()` (load persisted features) ? re-score each variant ? `calibration.improvement_report()`
+ `calibration.leave_one_video_out()` + `metrics.score()`, graded vs golden (`gt_loader.load_gt_intervals`).
Reports per-video **and** LOVO-honest aggregate IoU/P/R/F1 deltas. **This is the A/B test, and it needs
almost no new scoring code** ? only the glue.

Honest reporting (default ON, additive): `compare(..., honest_stats=True)` also attaches a `"honest"`
block ? per-metric bootstrap **CIs**, a paired **?F1 CI + exact sign test** (C1), and **F1@k +
segment-count ratio** (C4) ? *alongside* the bare-mean fields (existing output unchanged; `honest_stats=False`
restores the legacy shape). A bare LOVO mean at n?6 is not promotable on its own; the lift is real when the
?F1 CI clears 0 **and** the sign test agrees. Wiring of the course corrections in
ANALYZERS_AND_RECONCILERS.md §13/C1+C4 (helpers: `backend/eval/eval_stats.py`,
`segmentation_metrics.py`); `record_experiment` carries the honest ?F1 into the leaderboard.

5.2 Shadow (for cues that touch detection, e.g. the person detector)
The new cue runs and **persists its signal** into the candidate `stats`/`detection_params`, but the
served decision stays control. Offline you then ask "what would variant B have scored?" via §5.1 ?
B never affects output until it wins. This is how a detection-touching change (not just a re-scoring
change) earns its way in without risk.

5.3 Promotion gate (no-regression, CI-enforced)
`regression.regress()` + `baseline.json` + `run_regression.py` exit codes. A variant becomes the new
default **only if** it doesn't regress the golden set beyond `tolerance` (exit 1 blocks). Control is
pinned ? **rollback = flip `default_segmenter`/variant config, never a code revert.**

---

6. No-regression guarantees (the heart of the owner's ask)

- **New cues/signals default OFF** ? per-cue enable flag in `IndexingConfig`; merged code can't change behavior until explicitly enabled.
- **Control variant pinned** ? frozen recipe + classifier model hash, always registered & default.
- **Promotion only via the eval gate** ? `run_regression.py` in CI (exit 1 blocks); no silent default change.
- **Experiments are records** ? variant-spec hash ? per-video + LOVO scores + label-set hash + timestamp, persisted to a small JSON **experiment leaderboard** (generalizes `baseline.json`). Answers "which signals ever helped, on which video slices," and makes every result reproducible.
- **Decoupled events** ? "merge an experiment" and "change the default" are separate, independently revertible actions.

---

7. What to build later (gap list ? thin, additive, owner-gated)

1. ~~**`backend/eval/experiment.py`** ? a `Variant` dataclass + `compare(videos, a, b)` composing the §3 functions (no new scoring math).~~ **DONE (2026-06-13).** Shipped `Variant` (+ pinned `CONTROL`), `compare` (labeled A/B with LOVO-honest learned-variant training, mirroring `distill_local`), `compare_unlabeled` (the **SxS-on-new-videos** mode ? agreed / A-only / B-only via `match_intervals`, no GT), `record_experiment` (? `eval_baselines/experiment_leaderboard.json`), and `load_video_candidates` (the `query_candidate_segments` bridge). JSON variant recipes first as recommended; a `VariantRegistry` only if the count grows. 19 tests incl. the no-regression invariant.
2. **Persist per-cue features** into candidate `stats` (person: `players_in_court`, `two_sided_motion`, `in_court_ratio`, `shuttle_player_colocation`; shuttle-state: `net_crossings`, `direction_reversals`; audio: hit times) so any future variant re-scores offline ? ties to `MULTI_SIGNAL_FUSION_PLAN.md` §3.2/§4.
3. **Per-cue enable flags** in `IndexingConfig` (default `False`); `default_segmenter` already exists. Add the `min_crossings` knob (fusion P2) here too.
4. **CI promotion lane** using `run_regression.py` against the golden `baseline.json`; write the experiment-leaderboard JSON.

---

8. Phasing (owner-gated; ONE PR per slice, off `master`)

- **P0 ? this doc**, indexed in `docs/README.md`, cross-linked from `MULTI_SIGNAL_FUSION_PLAN.md` and `GOLDEN_SET_VENDORING.md`. *(this deliverable; docs-only, no code.)*
- **P1 ? `experiment.py` `compare()`** ? **DONE (2026-06-13):** thin wrapper over `calibration` + `metrics` + `query_candidate_segments`; pure-logic, fully tested in this dev container (`compare`, `compare_unlabeled`, `record_experiment`, the no-regression invariant). `compare_unlabeled` adds the SxS read on **new unlabeled footage** the owner asked for (run control + candidate on a fresh video,

diff where they agree/diverge ? two reels via the existing `VideoCompiler` is the GPU-side follow-on).
- **P2 ? per-cue enable flags + feature persistence** into candidate `stats` (default OFF) ? couples with fusion P1/P2.
- **P3 ? CI promotion gate**: `run_regression.py` lane on the golden `baseline.json`; experiment-leaderboard JSON.

---

9. Scope guardrail

Single-tenant / local-first (per `PLATFORM_ARCHITECTURE.md`): a JSON experiment registry + the existing harness ? explicitly **NOT** MLflow / W&B / a service. Keep it the size of `baseline.json`.

---

10. Verification

- **Offline, runnable in this dev container (no GPU/torch needed for the glue):** unit test that `compare()` is deterministic on synthetic candidate rows; a **no-regression invariant test** asserting "all new-cue flags OFF ? predictions byte-identical to control"; a CI-gate test asserting `run_regression.py` exit codes (0 ok / 1 regress / 3 no-baseline).
- **Machine-side (owner, GPU/WSL):** persist candidates once on a real clip, run `compare(control, fusion)` LOVO, confirm the reported lift matches a manual `metrics.score()` spot-check.
- **Docs (P0):** links resolve; indexed in `docs/README.md`; cross-linked from `MULTI_SIGNAL_FUSION_PLAN.md` and `GOLDEN_SET_VENDORING.md`.

</content>

16. user (2026-06-21T13:56:39.818Z)

Golden Regression Fixtures ? video-free, non-attributable regression suite

> **Status:** `IN PROGRESS` ? owner-gated · **Owner:** Avi · **Created:** 2026-06-16 · **Last updated:** 2026-06-16 · **Legal:** owner **accepted the researched risk** 2026-06-16 (conditions §3 stand; this is the owner's informed acceptance, **not** a retained-counsel opinion)
> **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE, per archive policy)
> **Tracking anchors:** §7 progress tracker is the source of truth; indexed in `docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once work starts).
> **Relation to existing docs:** EXTENDS (does not supersede) `docs/GOLDEN_DATA_SHARING.md` (#175); reuses the eval stack documented in `docs/EVALUATION.md` / `docs/EXPERIMENT_HARNESS.md`; quality floor ties to `docs/QUALITY_ITERATIONS.md`.
> **Honesty note:** §3 (Legal) is **researched general information, NOT legal advice** ? it gates on retained counsel (§8). §2/§4/§7 mark which claims are **[verified]** against code vs **[design]** proposals.

---

0. TL;DR

Let a contributor on **another machine with no video, GPU, WSL, or DB** run the rally-segmentation regression suite from **committed structured files**, so refactors and non-quality PRs get a red/green side-effect signal. Two layers, built once and grown by accretion:

1. **Mechanics** ? freeze the deterministic pipeline at the **post-detector trajectory

seam** and replay it through the *already-shipped* eval stack (`experiment.py` / `metrics.py` / `windowing.py` / `regression.py`), with two auto-discovered gates: a **characterization/snapshot** gate (catches any behaviour change) and a **quality-floor** gate (catches quality drops).
2. **Privacy/legal filter (SEALED-FIXTURES)** ? only **de-attributed, owner-source, biometric-free** numbers are ever committed; richer/sensitive streams stay key-gated or out-of-repo. Non-attributability comes from **deletion-at-source**, not encryption. ? "Minimized" means **stream minimization** (shuttle-only, no person/audio/pose) ? **NOT** reduction of the shuttle stream to ~5 scalars: the de-attributed per-frame `trajectory.csv` IS committed (see §4c + its risk-acceptance re-confirmation note).

The convergence that makes this clean: the *privacy* answer (publish shuttle-only, drop person/pose streams) and the *determinism* answer (freeze only the `CONTROL` path ? which runs **no numpy execution**, keeping the numpy/BLAS learned scorer off the cross-machine gate) point at the **same minimal public tier**. That coincidence is the spine of the design. (numpy is still *imported* transitively via `experiment` and must be installed; the CONTROL replay just never calls into it.)

Two findings already actionable, independent of the rest:
- ? **Live de-attribution bug** *(verified)*: real player/venue names are committed today in `eval_baselines/heuristic_lovo_n6.json`, `eval_baselines/gemini_wrapper_2026-06-10.json`, and `backend/eval/eval_partial_labels.py` (the last also has a personal `C:/Users/avidu/...` path) ? and in git history. ? **Deliverable P0.**
- ? **Determinism trap** *(verified, corroborated by two independent red-teams)*: the learned variants (`shuttle_only`, `fusion`) train a numpy/OpenBLAS `DYNAMIC_ARCH` logistic regression whose decisions flip near the 0.5 threshold across CPUs ? would false-RED on a contributor's machine. ? the cross-machine gate must be **CONTROL/static-only**.

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1. Problem & goal

The rally-quality measurement is split across two machines by capability (`docs/GOLDEN_DATA_SHARING.md`):
- `backend/eval/fusion_golden.py` **extracts** `*_golden_features.json` ? needs the **video + trajectory CSV + labels + GPU/WSL** (GPU box only).
- `backend/eval/fusion_compare.py` / `backend/eval/ablation.py` **re-score** those JSONs on **pure CPU, no video** ? any machine.

The CPU re-scoring is *already* video-free, but the inputs live in gitignored `output/` / private OneDrive, so the regression tests that depend on them **skip everywhere** (`tests/test_ablation.py::test_litmus_reproduces_gate_a`, `tests/test_fusion_compare.py`). The committed `eval_baselines/` baselines exist but `regression.py` still scores DB-stored predictions (needs a pipeline run on the video).

Goal: commit a tiny, version-tagged, **numeric-only, non-attributable** per-fixture corpus + auto-discovered gates so a clean `git clone` / CI gets a real red/green regression signal with **zero video/GPU/DB**, while never leaking attributable or sensitive data.

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2. Decisions locked

#	Decision	Source	Implication
D1	**Corpus is owner-recorded / owned**	owner answer 2026-06-16	Legal analysis takes **Fork A** (clean): no third-party copyright/ToS infringement; concern is trade-secret/moat + privacy. Strongest non-attributability (no public reference to correlate against).
D2	**Repo private now, may open-source later**	owner answer 2026-06-16	Design for **public-grade hygiene from day one** ? git history is permanent; nothing scrubable-later.
D3	**Adults, consent not formalized**	owner answer 2026-06-16	**Person/pose/gait streams stay OUT of the public tier** (DPDP/GDPR derived-personal-data risk without consent). Public tier = **shuttle + abstract intervals + non-biometric features only**.
D4	**Freeze seam = post-detector trajectory** *(design; SHIPPED differs ? see ?)*			

workflow-1 synthesis | Widest deterministic surface; reuse the shipped eval stack. ? **Correction (audit 2026-06-16):** the shipped freeze **commits the de-attributed per-frame `(Frame,Visibility,X,Y)` `trajectory.csv`** (shuttle-only, time-origin reset) ? it is **NOT** minimized to ~5 aggregate scalars. See §4c (and its risk-acceptance re-confirmation note). |
D5	**Cross-machine gate is CONTROL/static-only** *(design)*	both red-teams [verified cause]	Learned variants (numpy/BLAS) are **owner-box/Tier-1 only**, asserted with tolerance, never as a cross-machine byte-snapshot.
D6	**Non-attributability = deletion-at-source, encryption = public-only barrier** *(design)*	workflow-2 synthesis	A contributor who runs the tests necessarily reads plaintext+key (DRM/analog-hole). So we **delete** attribution, not hide it. Encryption only stops the no-key public.
D7	**Two tiers** *(design)*	workflow-2 synthesis	**Tier-0** public/no-key (shuttle-only, de-attributed). **Tier-1** key-gated/out-of-repo (person/audio, full corpus), **skips cleanly when absent**.

---

3. Legal foundation (researched ? gates on counsel, §8)

Core verdict (settled across IN/US/EU): the derived numbers ? per-frame shuttle (x,y), rally start/end intervals, audio/court/aggregate-person feature values ? **do NOT inherit the source video's copyright.** They are uncopyrightable *facts/measurements*, not authored expression, and not a §101 "derivative work."

Jurisdiction	Copyright of the data	Database right	Contract / ToS	Net
India (primary)	Not copyrightable as facts (*EBC v. D.B. Modak*); but compilations get thin protection on a low "skill/labour" bar (*Burlington*, *OLX v. Padawan*) ? only bites if **lifted from a 3rd-party dataset**, not self-derived	**None** (no sui-generis right) ? a real permissiveness edge	Anti-scraping ToS (untested in court) + **IT Act s.43/s.66** civil/criminal exposure for 3rd-party ingestion	**Owned-source + minors-excluded + no person/pose = clean**
US	Not copyrightable (*Feist*; §102(b); *NBA v. Motorola*)	None (*Feist* killed sweat-of-the-brow)	**Dominant risk.** ToS enforceable where copyright/CFAA fail (*ProCD*, *hiQ* ? injunction + **forced deletion**). Remedy for redistribution is takedown, not nominal damages	Publishable as facts **iff** owned/permissioned source + no barring ToS + no person/biometric/minor stream
EU / UK	Not copyrightable (*Football Dataco v. Yahoo*)	**Genuinely contestable.** *Football Dataco v. Sportradar*: recording pre-existing on-court facts = "**obtaining**", not "creating" ? our "we created it" premise is **overconfident**; harm test = *CV-Online Latvia*	Ryanair v. PR Aviation: ToS can ban extraction/redistribution even with no IP right	Get EU+UK advice before redistributing numeric streams at scale

Provenance is the decisive fork. **Fork A (owner-recorded, unpublished)** ? the only path safe to commit publicly; collapses to a trade-secret *choice* (Indian breach-of-confidence / Contract Act 1872 / IT Act s.72 ? **not** US DTSA) + privacy. **Fork B (broadcast/YouTube/scraped)** ? do **not** commit; the extraction act needs a license or now-**contested** US fair use (*Thomson Reuters v. Ross* 2025; USCO May-2025), and unlawful acquisition independently sinks it (*Bartz v. Anthropic*, ~$1.5B settlement). Caveat: owner footage **shot at a third party's event** (tournament/club) carries ticket/venue/organiser/broadcaster terms ? treat as Fork B until cleared.

Publish-by-stream (privacy):
- ? **Publish-OK:** shuttle (x,y)+Visibility (inanimate object), abstract rally/GT **intervals** (identity-stripped), non-biometric audio/court/aggregate-person features detached from identity.
- ?? **Conditional (effectively never in v1):** person bounding boxes that *persistently single out a player* ? flips to GDPR Art.9 the moment it identifies. This architecture is prone to that flip ? treat per-player tracking as a tripwire.
- ? **Never publish:** pose/skeleton/**gait** (re-identifies at 80?95% even with faces blurred ? *Weiss et al. 2025*), face crops, any named-player?movement mapping. Matches the existing "gait/biometrics strictly future (GDPR Art.9)" guardrail.
- ? **Minors flip everything:** DPDP "child" = under 18; s.9(3) **prohibits behavioural monitoring** (the Rule-12 exemptions don't cover this project). A persistent per-player signature on a minor is, conservatively, prohibited. **Exclude minors** from any persisted

stream beyond inanimate shuttle/interval data.
69
70 **Non-attributability is relative, not absolute.** A per-video trajectory is a
fingerprint (de Montjoye: 4 points ? 95% re-ID). It's only non-attributable because the **Fork-A
source stays unpublished** (no public reference to correlate against). Per *EDPS v. SRB* (CJEU
C-413/23 P, 2025) it **remains personal data to the holder** who can re-run the detector. So:
conditional, revocable (publish the source and the link snaps back), forward-looking (re-assess
as the public-video environment grows). Never market it as "anonymous."
71
72 *Key authorities:* Feist 499 U.S. 340; NBA v. Motorola 105 F.3d 841; 17 U.S.C.
§101/§102(b); Sega v. Accolade 977 F.2d 1510; Football Dataco v. Yahoo (C-604/10) & v.
Sportradar (C-173/11); BHB v. William Hill (C-203/02); CV-Online Latvia (C-762/19); Ryanair v.
PR Aviation (C-30/14); Thomson Reuters v. Ross (D. Del. 2025); EBC v. D.B. Modak; DPDP Act 2023
s.3/s.9 + Rules 2025; EDPS v. SRB (C-413/23 P). Full URLs in the research artifact (§10).
73
74 ---
75
76 ## 4. Architecture (design)
77
78 ### 4a. Freeze seam + reuse map
79 Freeze at the **trajectory CSV ? CONTROL-path replay**. One shared helper
`backend/eval/golden_fixtures.py::replay_fixture()` is the *single* code path used by both the
freeze tool and both gates. As **shipped**, that path is (per the module docstring):
80 `parse_trajectory_csv ? distill_local.build_candidates(proposal=(vt,mg,mwd)) ?
VideoCandidates(5 shuttle cols only) ? experiment.predict(CONTROL) [keep all ? merge_overlaps] ?
metrics.score`.
81 Reuses **verbatim**: `metrics.score/interval_iou`,
`experiment.predict/CONTROL/VideoCandidates`, `distill_local.build_candidates`,
`tracknet_runner.parse_trajectory_csv`/`trajectory_to_action_windows`, `rally_gate` net-
crossings (inside `build_candidates`), `gt_loader.load_gt_intervals` (strict),
`regression.gt_hash`, `eval_baselines/` committed-baseline precedent.
82 > ? **Correction (audit 2026-06-16):** the replay does **NOT** import `windowing.py` ?
there is no `windowing.build_windows(preset)` / `WindowingPreset` / "default SERVED" call. The
windowing params travel as a **raw 3-tuple `(vt, mg, mwd)`** pinned in each fixture's
`provenance.json` and passed straight to `build_candidates`; the harness deliberately avoids the
named `windowing.SERVED` constant so a future SERVED retune can't silently churn fixtures.
(`segmentation_metrics`, `experiment.compare/Variant.spec_hash`, `regression.save_baseline`,
`eval_stats` are **not** on the M1 replay path either ? the quality-floor gate M4 will add the
baseline/stats pieces.)
83
84 ### 4b. Tiered model (SEALED-FIXTURES)
85 | Tier | Where | Key? | Contents | Test behaviour |
86 |---|---|---|---|---|
87 | **Tier-0** | `eval_baselines/fixtures/` (tracked) | none | opaque surrogate slug,
fps/frame_width (retained, NOT quantized), the **de-attributed per-frame `trajectory.csv`**
`(Frame,Visibility,X,Y)` (shuttle-only, time-origin reset), the 5 shuttle feature columns inside
`expected.json`, strict-slug `label`, relative `gts`. **No person/audio/pose.** ? NOT minimized
to 5 scalars ? see §4c. | **Unconditional** ? runs on every PR/fork; turns today's skipped
litmus into a real public gate |
88 | **Tier-1** | OneDrive (default) **or** SOPS-encrypted in-repo | token/age key | full
corpus **with** person/audio; authoritative ?F1/CI/sign-test + exact Gate-A litmus | **Skips
cleanly** when key/data absent (generalize `_golden_present()` ? `_tier1_present()`) |
89
90 A fresh no-key clone is always **green**; coverage never depends on a secret.
91
92 ### 4c. Anti-attribution measures (Tier-0)
93
94 > ? **RISK-ACCEPTANCE RE-CONFIRMATION REQUIRED (audit 2026-06-16).** An earlier draft of
this doc claimed Tier-0 commits "only ~5 aggregate shuttle scalars" and "**never** the per-frame
CSV". **That is not what ships.** The shipped freeze (`freeze_real_fixture`) **commits the FULL
per-frame `(Frame,Visibility,X,Y)` `trajectory.csv`** ? de-attributed (time-origin reset +
opaque random slug + metadata strip), but **not minimized to 5 scalars**. The per-frame stream
is the de Montjoye fingerprint (4 points ? 95% re-ID against a published reference). The owner's
2026-06-16 risk acceptance (§7 P2) was recorded against the inaccurate "5 scalars" basis and
MUST be RE-CONFIRMED on this corrected basis before any **real** Tier-0 fixture is committed

(P3). Synthetic fixtures (current state) carry zero provenance risk regardless.

- **Opaque RANDOM surrogate id** (`os.urandom` + a private surrogate?source map kept off-git) ? *not* the MD5 `video_id`, and *not* HMAC(salt, video_id?name) (red-team: brute-forceable over a tiny known roster if the salt leaks). Never persist the source `name`/filename.
- **Strict-slug `label` + dropped `source_note`** ? the operator `label` is validated against `^[a-z0-9][a-z0-9_\-]*$` (a free-text identifier like a player/event name is refused), and operator free-text `source_note` is **accepted-but-NOT-persisted** (never reaches the committed `provenance.json`) ? both close the free-text identifier leak `_scan_forbidden` could not police (audit fix #1).
- **Metadata strip** ? no filename, video_id, match/event/player identity, capture date.
- **Time-origin reset** ? subtract per-fixture `t0` (6dp-rounded) from all start/end/gts. **Metric-invariant within `_FLOAT_TOL`** (`metrics.interval_iou` and start/end-error are pure differences): the genuine (raw) reset-vs-no-reset score delta sits at the float-error floor (~3.3e-7); bit-identical for whole-second offsets. See §4c-note below and §6.
- **Per-frame trajectory IS committed (de-attributed), NOT minimized to scalars** ? Tier-0 commits the de-attributed per-frame `trajectory.csv` (shuttle stream only). It is **not** reduced to ~5 aggregate scalars; the 5 shuttle *feature* columns appear (computed) inside `expected.json`, but the raw per-frame `(Frame,Visibility,X,Y)` stream is part of the committed fixture. (This is the corrected basis the ? note above gates on.)
- **Stream minimization (streams, not resolution)** ? Tier-0 keeps the shuttle block only; person/audio excluded; pose/gait/face have **no field in the schema by design** (positive `_scan_forbidden` assertion on the written bytes).
- **fps/frame_width are RETAINED as scale constants** ? they are NOT quantized today (quantization is a deferred P3 option, see §7 ?). Reconcile any "quantize fps/frame_width" wording elsewhere to this.
- **Provenance hard-fail gate** ? the de-attribution pass refuses to emit a public fixture for any record not tagged owner-recorded(Fork-A) + minors-excluded + non-blank-license, and refuses a negative reset label (timeline predates the trajectory). Writes go to a temp dir and are promoted only after the privacy scan passes, so a refused freeze leaves no partial fixture (audit fixes #3a/#3b).

4d. The two gates
- **Characterization/snapshot** ? replay fixture, **parse-compare** JSON (never byte-compare ? CRLF/float-format safe; `core.autocrlf=true` here), exact on int/structural fields, `approx(abs=1e-6)` on floats; stamp + check windowing-preset/label/gt fingerprints (incommensurable ? loud fail). Catches any behaviour change even at constant F1. **CONTROL path only** across machines.
- **Quality-floor** ? score vs committed labels; assert per-video + mean F1 ? floor in `eval_baselines/regression_baseline.json` (created here; carries a **windowing fingerprint** so a served-default change is flagged incommensurable, not silently compared); emit `eval_stats` paired-delta-CI + sign test (NUMBERS INFORM); add a **metrics-vs-base-branch** paired delta. Synthetic-tier floors measure **correctness, not field quality** (documented).

4e. Serialization & optional protobuf
JSON with `sort_keys=True` + pre-quantized floats + LF is sufficient. **Protobuf is optional (O1)** ? adopt only if cross-OS byte-stable snapshots are wanted (proto3 `rally.fixture.v1`, codegen + `protoc && git diff --exit-code` guard, `*.pb binary -text`). Protobuf is **encoding, not secrecy** (trivially `--decode_raw`).

4f. Encryption & keys (Tier-1 only)
Default = out-of-repo + token (existing OneDrive `RALLY_GOLDEN_DIR` seam ? zero sensitive bytes in git history, sidesteps the legal "redistribution" act). **Alternative = SOPS + age** (per-recipient, revocable, value-level diffs) for Fork-A-clean-but-private data wanting in-repo versioning. Rejected: git-crypt (whole-repo re-key), git-lfs (storage ? secrecy). CI secret exposed **only to the Tier-1 job on trusted branches, never to fork-PR runners**. The **HMAC/surrogate salt** (if used) and the surrogate?source map live with the private corpus, rotated per release. Document explicitly that Tier-1 crypto defends **only** the no-key public.

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5. Threat model (who's protected vs not)

Actor	Protected against	NOT protected (by design / residual)

TM1 no-key public	map fixture?source/match/player; decrypt Tier-1; confirm a source even holding the candidate file	read Tier-0 numbers + run shuttle-only harness (intended; useless-in-isolation)
TM2 keyed contributor	recover filename/video_id/match/face/pose/gait ? *never written* (survives the analog hole)	read every value the suite reads (unpreventable; not pretended otherwise)
TM3 CI runner	leaked log can't leak attribution it never had	runs the suite (intended); Tier-1 secret never on fork runners
TM4 malicious keyed insider	correlation needs a public reference; Fork-A source is unpublished ? "not reasonably likely"	**residual:** conditional + revocable ? publish/leak the source later and the link can snap back (forward-looking re-test duty)
TM5 re-ID adversary	source unpublished + reset timeline ? no back-match against a public reference	? becomes live if a Fork-A source is ever published ? **and the committed per-frame trajectory IS a fingerprint** (de Montjoye 4-point re-ID), NOT just ~5 scalars (corrected §4c). The protection is the unpublished source, not minimization
TM6 accidental plaintext commit	required CI leak-scan rejects MD5/known-name/non-surrogate/plaintext-SOPS	if the required check is disabled, a leak lands

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6. Honest limits ? what a green suite does NOT prove

- Green proves only that the **deterministic post-trajectory transforms** are unchanged for the **frozen cases**. It does **not** cover: the WASB/TrackNet detector, `_probe_fps_width`/cv2 frame-seek, the AI/provider handoff, chunking/re-encode, the ffmpeg stitch, DB persistence/migrations, or the API filter. *A refactor that breaks decoding or the detector passes these gates green.* The owner-box `regression.py` DB run remains the only full-chain check.
- Characterization = **behaviour-locking, not correctness** ? it freezes current behaviour *including current bugs*. "Passing" = "behaviour unchanged", never "behaviour correct". This caveat must ride in the **test pass/fail message**, not only the docs.
- **Synthetic-tier floors = plumbing correctness, NOT field quality.** Never cite synthetic F1 in `QUALITY_ITERATIONS.md` / the paper as rally-detection performance (tag `kind: synthetic`).
- Tier-0 public coverage is **shuttle-only**; the person/audio fusion ?F1 and exact Gate-A values need Tier-1 (skipped on no-key clones).
- Data is **not "anonymous"** absolutely (EDPS v. SRB) ? relative, conditional, revocable, forward-looking.
- The committed Tier-0 stream is the **de-attributed per-frame `trajectory.csv`**, NOT ~5 scalars ? a per-frame trajectory is a de Montjoye fingerprint; the only protection is the **unpublished source** (§4c ?, §5 TM5).
- **Time-origin reset is metric-invariant within `_FLOAT_TOL`, not byte-exact for every offset.** The genuine (raw) reset-vs-no-reset score delta is at the float-error floor (~3.3e-7); it is **bit-identical for whole-second offsets**, but because 6dp-rounding is not additive (`round6(a)-round6(b) ? round6(a-b)`) the *serialized* 6dp boundary error can differ by one 6dp ULP (1e-6) on a sub-second offset ? the serialization grid, not a metric divergence. (Audit fix #2 rounds `t0`+labels to 6dp before the shift to keep the residual ? the grid step.)
- Encryption gives **no secrecy from a running contributor**; protobuf gives **no secrecy at all**.
- De-attribution does **not** cure provenance/licensing ? it sits *behind* the per-record provenance gate + counsel sign-off.
- `--refresh-snapshot` can rubber-stamp a real regression; mitigation is **review of the visible diff** + a separate, deliberate floor-baseline update. **A recorded `--reason` is REQUIRED** (the CLI errors without one ? audit fix #6).

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7. Deliverables & progress tracker ? **source of truth**

Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. One **small PR per row**, branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.

ID	Deliverable	Depends on	Counsel-gated?	Status	PR
P0	**Leak remediation + required CI leak-scan.** Scrub real names/paths in

`eval_baselines/heuristic_lovo_n6.json`, `gemini_wrapper_2026-06-10.json`, `backend/eval/eval_partial_labels.py`, `tests/test_cleanup_caches.py` ? opaque surrogate keys + private map; add CI leak-scan (MD5 / known-name / non-surrogate / plaintext-SOPS) as a **required status check**; decide on git history (PR #77, accept vs `filter-repo`). | ? | No (owned data hygiene) | ? | ? |

152 | **M1** | **Fixture framework + synthetic Tier-0 + shared replay helper** (also delivered M2-core + M3 ? see below). `backend/eval/golden_fixtures.py` (`replay_fixture`, schema dispatch, `Path(__file__)`-relative) + 3 **tool-generated** synthetic fixtures + manifest + README + `.gitattributes` LF pin + `tests/test_golden_fixtures.py`. CONTROL-only (no numpy execution on the replay path); cross-OS verified. | ? | No (synthetic) | ? | [#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) |

153 | **M2** | **Freeze/refresh extractor.** Core (`freeze_synthetic` / `refresh_snapshot` / `--check` CLI + idempotence test) shipped in M1 (#189); the `--license`/`--minors-free`/`--owned` **refusal gate** shipped with P1 (this PR). | M1 | No | ? | [#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) · _this PR_ |

154 | **M3** | **Characterization gate** ? `tests/test_golden_fixtures.py`: parse-compared (exact-int / approx-float 1e-6), **CONTROL-only**, positive no-person/audio assertion, perturbation + idempotence. **Shipped in M1 (#189).** (Windowing/label fingerprint guard + row-count cap ? fold into M4.) | M1 | No | ? | [#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) |

155 | **MX** | **Cross-OS determinism guard (CI).** Lightweight `golden-crossos` CI job (ubuntu/windows/macos) running `--check` + the fixtures test ? locks the cross-OS guarantee (CONTROL replay is numpy-only, no GPU stack; a *separate fast job*, not full-suite×3). | M1 | No | ? | [#191](https://github.com/avidullu/badminton-highlight-indexer/pull/191) |

156 | **M4** | **Quality-floor gate + first `regression_baseline.json`.** `tests/test_golden_quality_floor.py`; create `eval_baselines/regression_baseline.json` with **windowing fingerprint**; metrics-vs-base paired-delta; corpus-level (not per-slug) LOVO for runtime. | M1, M2 | No | ? | ? |

157 | **P1** | **De-attribution + minimization** ? `golden_fixtures.freeze_real_fixture()` (extends the M1 module, not a separate script): **opaque random `fx_<hex>` surrogate slug**, metadata strip, metric-invariant **time-origin reset**, strict GT loader, **positive no-person/audio/MD5/source-path assertion**, manifest non-clobber; `--freeze-real` CLI with the **refusal gate** (exit 2 unless owned + minors-free + license). Tooling-only, 12 synthetic tests; **no real data committed**. | M1 | No (tooling) | ? | _this PR_ |

158 | **P2** | **Owner-accepted (researched-risk), 2026-06-16.** Owner accepted the researched IP/privacy analysis (§3) to proceed **under its conditions** (owned/Fork-A source · minors excluded · biometric/person streams excluded · de-attributed). Owner's informed risk acceptance ? **NOT a retained-counsel opinion**; optional future counsel confirmation per §8. | P1 | Owner-accepted | ? | ? |

159 | **P3** | **Tier-0 REAL subset.** Commit de-attributed, minors/biometric-excluded, owned Fork-A real fixtures (unblocked by P2). Hard-gated **in code** by P1's refusal gate (owned + minors-free + license required) ? needs the owner's per-video clearance list. | P1, P2 | conditions §3 | ? | ? |

160 | **O1** | *(optional)* **Protobuf + deterministic serialization.** `rally.fixture.v1` proto3, codegen + `protoc`/`git diff --exit-code` guard, canonical writer, `.gitattributes` binary. Only if cross-OS byte-stable snapshots are wanted. | M1 | No | ? | ? |

161 | **O2** | *(optional)* **Tier-1 key-gated path.** `scripts/fetch_golden.py` (OneDrive token, default) and/or SOPS+age; `_tier1_present()`; `RALLY_GOLDEN_DIR` wired; Tier-1 CI job gated on secret, never on fork runners. | M1 | No | ? | ? |

162 | **DOC** | **Discoverability cross-links** ? added the `eval_baselines/fixtures/` row to `CODE_MAP §0` + a `golden_fixtures.py` entry to `CODE_MAP §2.3`; cross-linked `GOLDEN_DATA_SHARING.md` (committed subset ? private OneDrive corpus). README indexed in #186; memory `current-state` kept live. (EVALUATION/QUALITY_ITERATIONS cross-links **deferred** ? that doc surface is hot in the concurrent quality track; avoid collision.) | M1 | No | ? | _this PR_ |

163

164 **Must-fix-before-any-public-commit (red-team), folded into the rows above:** ? numeric-tolerance not byte-equality + multi-OS matrix + CONTROL-only (M3/D5/MX ?); ? scrub names in **all** files + history decision (P0); ? leak-scan a **required** check, hooks are convenience only (P0); ? person-optional loader + **positive** no-person assertion, don't trust the silent `0.0` default (P1 ?); ? random surrogate id not HMAC-over-roster (P1 ?); ? quantize/drop fps/frame_width (P1 ? fps/fw retained as scale constants; quantize is a P3 option); ? document Tier-0 omits the fusion path (DOC); ? counsel + Fork-A + minors/biometric exclusion as hard preconditions (P2 owner-accepted; enforced in code by P1's refusal gate ?).

165

Progress log (quantitative ? coverage must hold/improve, CI incl. the 3-OS cross-OS job green)
PR	Deliverable(s)	Suite	Coverage	Notes
[#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189)	M1 + M2-core + M3	746 pass / 7 skip	80.41%	synthetic Tier-0 + CONTROL replay + characterization; cross-OS proven (CI Linux replayed Windows-frozen fixtures)
[#191](https://github.com/avidullu/badminton-highlight-indexer/pull/191)	MX + tracker/legal reconcile	full suite green	80.41%	`golden-crossos` CI job green on ubuntu+windows+macos ? cross-OS **enforced**; legal ? owner-accepted
this PR	P1 + M2 refusal gate	759 pass / 7 skip	**80.49%** (+0.08)	`freeze_real_fixture` + refusal gate; **+13 tests**; tooling-only, no real data committed
this PR (audit-hardening)	adversarial-audit fixes #1?#17	784 pass / 7 skip	**80.71%** (+0.22)	strict-slug `label` + dropped `source_note` (#1); 6dp-rounded reset within `_FLOAT_TOL` (#2); temp-dir-then-promote + negative-label refusal (#3); exact-membership source-path scan (#4); key-set-equality in `compare_expected` (#5); `--reason` mandatory (#6); independent schema guard (#7); `max_win` recorded-not-applied note (#8); numpy-execution wording (#9); +failure-semantics & by-kind tests (#10/#11); `test_golden_real_fixtures.py` added to `golden-crossos` CI (#12); doc reconciled to shipped reality incl. per-frame-CSV honesty + risk re-confirm (#13?#17). Synthetic `--check` byte-identical (unchanged).

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8. Open owner / counsel questions

Counsel questions ? documented basis (the owner accepted the researched risk on 2026-06-16; see P2). These remain the record, and a retained-counsel confirmation is optional/recommended before scaling beyond a small owned, minors-free, de-attributed subset:
1. Does the EU/UK **sui-generis database right** subsist in our self-derived shuttle/rally streams on the *Sportradar* "obtained vs created" reasoning, and does redistributing a de-attributed Tier-0 subset cause *CV-Online* harm?
2. Post *Thomson Reuters v. Ross* + USCO-2025, is **US fair use for the extraction step** defensible if the dataset could compete in a "data/analytics" market ? does publishing only numbers (never footage) matter?
3. India: are anti-scraping ToS enforceable; **IT Act s.43/s.66** exposure for any 3rd-party ingestion; does pending *ANI v. OpenAI* (~Apr 2026) move it? Confirm **DPDP s.9/Rule 10** and whether per-player tracking is prohibited s.9(3) monitoring of a minor.
4. Sign-off on the **final committed Tier-0 subset + provenance log**; is non-attributability-via-private-source a defensible posture to assert given its conditional/revocable nature?

Owner (strategic):
5. Publishing the Fork-A subset **forfeits the trade-secret moat** (Indian breach-of-confidence framing) ? weighed against community/benchmark/paper aims. Your call.
6. **Forward-looking commitment:** do not publish a highlight reel of the *same* source whose fixtures are committed (without re-clearing), and accept a periodic re-assessment as the public-video environment grows.
7. **Scope of first public commit:** synthetic-only (zero provenance risk, recommended) vs a counsel-cleared real subset.
8. **Tier-1 mechanism:** out-of-repo+token (recommended) vs SOPS+age in-repo. And: is protobuf (O1) wanted, or is canonical-JSON enough?

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9. Definition of done

The project is **DONE** (flip `Status ? DONE`, archive) when:
- [ ] **P0, M1?M4, P1, DOC** are ? and merged.
- [ ] A clean `git clone` on **win + ubuntu + mac** runs `python run_all_tests.py` and the **Tier-0 gates pass green with no key / no video / no GPU / no DB**.
- [ ] The **CI leak-scan is a required status check** and passes (no MD5 / real-name / non-surrogate / plaintext-SOPS in the tree).
- [ ] Characterization + quality-floor gates are **parse-compared, CONTROL-only, windowing-fingerprinted**, and demonstrably catch a seeded behaviour change.
- [ ] `docs/CODE_MAP.md §0` + TEST MAP updated; this doc cross-linked; `NEXT_STEPS.md` +

memory `current-state` point here.
- [ ] **Either** the public commit is synthetic-only, **or** (**P2 owner-accepted ? + P3**) for a real-data commit, under the §3 conditions enforced **in code** by P1's refusal gate.
- [ ] §6 honest-limits caveat rides in the test pass/fail messages, not only the docs.

Optional O1/O2 may remain ? without blocking DONE (they're earned, not assumed).

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10. References

Internal code anchors (verified): `backend/eval/{regression.py,fusion_golden.py,fusion_compare.py,ablation.py,experiment.py,metrics.py,windowing.py,distill_local.py,gt_loader.py,eval_stats.py}` · `backend/pipeline/detectors/{base.py,tracknet_runner.py,rally_gate.py}` · `backend/utils/hashing.py::compute_video_id` · `eval_baselines/{heuristic_lovo_n6.json,gemini_wrapper_2026-06-10.json,experiment_leaderboard.json}` · `tests/test_ablation.py::test_litmus_reproduces_gate_a` · `.github/workflows/ci.yml`.
Internal docs: `GOLDEN_DATA_SHARING.md` (#175), `EVALUATION.md`, `EXPERIMENT_HARNESS.md`, `QUALITY_ITERATIONS.md`, `MULTI_SIGNAL_FUSION_PLAN.md`, `CODE_MAP.md §0`.
Legal authorities: see §3 (full URLs in the research workflow artifact). *This section is researched general information, not legal advice.*

Changelog
- **2026-06-16** ? Initial draft synthesized from two multi-agent workflows (mechanics design + 2 critiques; legal memo IN/US/EU + SEALED-FIXTURES architecture + red-team). Verified the live name-leak and the BLAS-determinism cause against code. Status: DRAFT, owner-gated.

17. assistant (2026-06-21T13:56:44.132Z)

Now let me read the ROORA labeling guidelines (it's in archives per the cross-links) and check QUALITY_ITERATIONS for the latest measured state to validate tracker freshness.

18. user (2026-06-21T13:56:46.433Z)

GOLDEN_SET_VENDORING.md
README.md
ROORA_LABELING_GUIDELINES.md
---QUALITY ITER head---

Rally-detection quality iterations ? living log

Purpose. A running, reverse-chronological log of how we evolve rally-detection quality: each iteration's hypothesis, what shipped, the measured (or expected) effect, and the lesson. This is the
live continuation of the historical trail in
[`archives/quality-iterations/`](archives/quality-iterations/README.md) ? per the archive policy (`CLAUDE.md`), **terminal-state** (DONE/REJECTED) work gets a dated subdir there; this file stays at
`docs/` top level and keeps growing, then its entries graduate to the archive when they conclude.

Methodology (non-negotiable). Every quality change rides the experiment harness (EXPERIMENT_HARNESS.md): new cues land **flag-gated, default OFF**; they're
graded **offline against golden labels** (per-video + LOVO-honest); promotion to default happens **only
on measured lift** through the `run_regression.py` gate; **rollback = flip a config flag, never `git revert`.** Cues are designed in MULTI_SIGNAL_FUSION_PLAN.md; scoring
in EVALUATION.md; the 2-layer model in
HOW_RALLY_DETECTION_WORKS.md.

Golden corpus = the 15 labelled matches in `output/human_lovo_manifest.json` (the original 6
?
`adarsh_avi_singles, kushagra_singles, largetest_doubles, gbaaddy, testlarge_short,
mahadevpura_singles` ? owner-confirmed 2026-06-13, since grown to 15 with the later
GoPro/Boxhill clips). The model trained on these is what we run alongside the existing
detector on NEW footage for the side-by-side review.

Baselines that drive strategy (IoU 0.5 vs human golden labels)

Path	Mahadevpura (clean-ish)	Kushagra (multi-court)
Heuristic (velocity windowing)	0.657	0.157
Gen-0 owned head (LOVO, n=6)	0.744	0.561
Two-stage WASB+Gemini refine	0.83	?
Gemini wrapper (`gemini-flash-latest`)	0.93	0.66

Reading: clean footage is commoditized (serve via Gemini); **multi-court drops the commodity
wrapper
to 0.66 ? that is the owned model's ship target.** Baselines tracked in `eval_baselines/`. The
owner's
observed false positive ? a player **holding/flicking** the shuttle between points counted as a
rally ?
is the precision bug the fusion gates target.

Current track (live) ? newest first

P7 ? Long-rally (>5s) quality lens (eval) + reel-selection knob ? MEASURED, default-OFF *(2026-06-18)*

Hypothesis (owner). Local OVER-FIRES, and its false positives are mostly SHORT (motion
blips,
background-court fragments on multicourt). Filtering the eval to **long rallies (>5s)** ? the
eye-catching highlights that carry the product ? should strip most short FPs and reveal local's
real
long-rally quality, answering the North-star "do we beat Gemini on the single-match long-rally
happy
path?". **Duration** (end?start), NOT shot-count, is the filter: golden `shots_count` is blank
and the
no-AI path reports it "Unknown", whereas duration comes from the start/end labels we already
trust.

What shipped (both default-OFF, zero behaviour change at the defaults).
- **Eval lens** ? `rally_seg_eval.py --min-duration <s>` filters BOTH predictions AND golden GTs
to
  rallies longer than `s` *before* matching, consistently across the tIoU and R2 tolerance paths
(a
  FILTER, not a new metric). Plus `--stratify` / `--long-tau` (default 5s): a `{all vs long} ×
{single
  vs multicourt}` aggregate table. Court type = the committed `MULTICOURT_CLIPS` set (the
manifest is
  gitignored, so the set is the auditable source), overridable by a manifest `court_type` field;
kept
  in sync with [`GOLDEN_VIDEOS.md`](data_pipeline/GOLDEN_VIDEOS.md).
- **Product knob** ? `stitching.min_rally_duration` (s, 0 = OFF) drops short rally pairs before
the
  reel compile in `tools/generate_highlights.py` ? a cleaner reel-selection filter than
  `min_shot_count` (which leans on the "Unknown" shot_count).

**Measured (n=15 golden, milestone cap6+R4, LOCAL vs golden; reproduce: `python -m
backend.eval.rally_seg_eval --stratify`):**

| stratum | court | n | tolF1 | tolP | tolR | tIoU F1@0.5 | seg_ratio |

all	single	11	0.375	0.326	0.464	0.482	1.434
all	multicourt	4	0.294	0.238	0.393	0.430	1.601
all	all	15	0.353	0.302	0.445	0.468	1.478
long>5s	single	11	0.264	0.278	0.267	0.416	**1.037**
long>5s	multicourt	4	0.240	0.197	0.310	0.430	**1.597**
long>5s	all	15	0.257	0.256	0.279	0.420	1.186

Pooled counts: single **608 preds / 416 GTs** (only 36% of preds are >5s vs 50% of GTs ? long pred:GT ? **218:210 = 1.04**); multicourt **288 preds / 179 GTs** (55% of preds >5s ? long pred:GT = **158:100 = 1.58**).

Findings ? the lens turned a one-line hypothesis into a 3-part measured map (NUMBERS INFORM):
1. ? **Single court, count-level: thesis HOLDS.** Local's over-production is dominated by SHORT windows
   ? restricting to >5s collapses seg_ratio **1.43 ? 1.04** (preds ? GTs in count). The short-blip FPs
   are real and duration-filterable.

19. assistant (2026-06-21T13:56:57.690Z)

This confirms the corpus has grown to 15 (the §7 tracker in RALLY_QUALITY_RESEARCH still says n=6 in places). Let me read the ROORA guidelines briefly and check the rest of QUALITY_ITERATIONS to capture the latest measured state and any newer items beyond the tracker.

20. user (2026-06-21T13:56:58.636Z)

```
# Roora Annotation Guidelines ? Rally Golden Set

**Version: v8 (2026-06-13).** Supersedes the older Drive copies (v3/v5/v6/v7) **and** the internal
"DRAFT v2" label ? those numbers never matched each other; **v8 is the single label to use with
Roora going forward.** Content is otherwise the finalized 2026-06-12 guideline plus the
fusion-driven additions marked **`? ADDED (fusion)`** below.

> **?? Canonical home / provenance.** **This in-repo file is the sole canonical copy of the guideline**
> ? it lives here because the multi-signal-fusion features that motivate the new fields (court
> calibration, singles/doubles) originate in *this* repo (see
> [MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md) and
> [GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md), ADR-006). The annotation deliverable still imports
> into the sibling **`sports-data-collector`** repo via `python -m src.cli.curate import-local`; if a
> mirror is ever staged there for the collector's intake docs, it must be regenerated from this file
> (and the `? ADDED (fusion)` callouts are exactly the deltas to port).
>
> **`TODO(avidullu)` markers are preserved** ? they are owner-only items (Drive links, intake
> manifest, NDA counsel review) and must be filled before sending to Roora. Search for `TODO(avidullu)`.

Brief (vendor spec) + Illustrated Guide (rater how-to), in one doc.

---

# PART A ? Roora Annotation Brief (vendor-facing spec)

**Project:** racket-sport rally segmentation golden / regression dataset.
```

Sports in scope: Badminton, Pickleball, Table Tennis (each in singles AND doubles).

This brief is the vendor-facing spec. The annotation deliverable is a per-rally CSV that imports
directly via `python -m src.cli.curate import-local`.

0. What we need in one sentence

For each provided match video, produce ONE CSV file listing EVERY rally with its start time, end
time, number of shots, and how it ended ? EXCLUDING all dead time.

1. Deliverable format (canonical)

One CSV per video (UTF-8), named exactly `<video_id>.csv` (we supply the `video_id`). One row per
rally, chronological, header row required.

Columns:

Column	Type / values	Meaning
`rally_number`	integer, 1-based, chronological ? no gaps, no repeats	rally index
`start_mmss`	`m:ss.mmm` (e.g. `1:12.40`)	rater-entered serve-contact time
`end_mmss`	`m:ss.mmm`	rater-entered ball/shuttle-dead time
`start_time`	decimal **seconds** (e.g. `72.40`) ? **AUTO-FILLED** from `start_mmss`	machine time
`end_time`	decimal **seconds** ? AUTO-FILLED from `end_mmss`	machine time
`shots_count`	integer	total racket/paddle/bat contacts (serve = shot 1)
`ending_reason`	`winner` | `forced_error` | `unforced_error` | `service_fault` | `let` | `other`	WHY it ended
`last_shot_outcome`	`in` | `net` | `out` | `n_a`	WHERE the last shot landed (`n_a` for service_fault/let)
`score_before`	optional, `"A-B"` ? blank if not confidently readable	score before
`score_after`	optional, `"A-B"`	score after
`notes`	optional free text ? **REQUIRED when `ending_reason = other`**	edge-case note

Times (mm:ss ? decimal seconds): raters type only `start_mmss` / `end_mmss`. The template Google
Sheet fills `start_time` / `end_time` with this formula (shown for `start_time`, row 2):

21. user (2026-06-21T13:56:59.813Z)

   are real and duration-filterable.
2. ?? **Multicourt: thesis FAILS.** Even among >5s rallies local over-fires **1.58×** ? adjacent-court
   games are genuine LONG rallies ? long *spurious* windows. **A duration filter does NOT clean
   multicourt; that needs court-localization** (Gate-B / court mask ? gap-closure Lever C). So "FPs are
   mostly short" is a *single-court* statement.
3. ?? **Strict tolerance precision does NOT jump** (tolP single 0.326 ? 0.278) even where the count
   normalizes, because the >5s pred set is contaminated by **over-MERGED windows** (kushagra med 9.9s,
   mahadevpura_1's 63s blob, GX010141 14.8s) and cap=6 truncates genuinely-long rallies so the strict
   |?end| ? 1.5 s match fails (tIoU, forgiving, holds ~0.42; strict tol drops). The long-rally
   strict-eval happy path is gated by merge-control + the cap-vs-long tension, not just FP-stripping.
4. ? **Product knob is still a clean reel win.** A reel doesn't need strict boundary

tolerance ? an over-merged 15 s clip is a fine highlight, and dropping <5 s pairs removes the single-court short junk. Defaults OFF; owner reviews the numbers before any flip.

Gemini: not computable in-checkout ? the Boxhill gemini intervals live in the owner-side scorer's DB (not committed). The lens is metric-agnostic: feed Gemini's intervals through the same `filter_by_min_duration` to get its long-rally row (owner-side one-liner). From gap-closure §3, Gemini already fires fewer/longer windows with ~0 short FPs, so a long filter mostly affects LOCAL.

Lesson. A plausible product one-liner ("filter to long ? precision jumps") split into three measured truths once instrumented: single-court count ?, multicourt ? (needs a court signal), strict-precision ? (needs merge control). Default-OFF lens + knob; the owner decides any flip on the numbers. *(P7 in [`RALLY_DETECTION_GAP_CLOSURE.md`](RALLY_DETECTION_GAP_CLOSURE.md) §7.)*

R1 ? MILESTONE LAUNCHED (cap=6 + R4 flipped default-ON) ? re-verified at n=13 *(2026-06-18)*
The R1 milestone is **shipped** (PR #204): the W1 cap=6 + R4 inactivity-end defaults are now ON. Full quality-impact table on both rulers, the over-seg guard, and the honest caveats are recorded in the **[R1 Milestone Launch Report](R1_MILESTONE_LAUNCH_REPORT.md)** (per the Launch-Report convention).

Re-verified on the grown n=13 corpus (the 2 new Boxhill clips GX010137/GX010141) ? the milestone holds and the headline is *stronger* than at n=11:

transition	n=11 (filed)	**n=13 (launched)**
baseline ? milestone tolF1	?+0.232, 10/0, p=0.002	**?+0.222, 12/0/1, p=0.0005**
cap12 ? cap6 tolF1	?+0.108, 10/0, p=0.002	**?+0.110, 12/0/1, p=0.0005**
cap6 ? +R4 tolF1	?+0.047, 8/0, p=0.008	?+0.040, **9/1/3**, p=0.0215
net over-seg guard	PASS (mr 0.694?0.150)	**PASS** (mr 0.651?0.161, 0 regressions)

Both new clips improve (GX010137 0.448?0.685, GX010141 0.333?0.431); tIoU ladder baseline 0.236 ? milestone 0.474. Independently reproduced by a 2nd harness (#214 `lovo_resweep.py`, matches to 3 dp; LOVO picks cap=6 **13/13** on tIoU) and survived a 5-lens adversarial workflow (every LOO fold clears the bar; worst-fold drop GX010128 still ?+0.202, p=0.0010 ? no single clip drives it). **Honest notes:** W1 cap is the dominant lever / R4 is the marginal +0.040 increment (1 loss = GX010137, a tolerance artifact); the gate is the NET rate-based guard (strict=False by design); GX010141 is the worst-behaved clip (future over-merge target). The nightly self-flags STALE post-flip ? re-baseline once with `--update-baseline`. W2 / hard-cap / R5 stay default-OFF.

R5 ? blob-fallback for the continuously-active blob (default-OFF, stratum tool) *(2026-06-17)*
The one clip the cap6+R4 milestone still fails is the **continuously-active blob** (mahadevpura_1: ~174 s of play with no inactivity gap to split on). The W1 gap-split finds no gap; R4 finds no slow tail; the result is one ~66 s mega-window that matches nothing (**tolF1 0.000**, the

degenerate 0-match outcome [`REAL_FOOTAGE_VALIDATION_REPORT.md`](REAL_FOOTAGE_VALIDATION_REPORT.md) §5 R5 names).

R5 (`window_blob_fallback`, default-OFF) is the last resort: **after** the gap-split and the R4 trim have each had their chance, any group still longer than `window_blob_factor` × the cap is a genuine blob ? midpoint-chop it to ? the cap, **log the forced cut**, and **flag every resulting window `forced_cut=True`** (surfacing the clip as one that needs rally-STATE cues ? net-crossings / two-sided motion ? not a bigger cap). `_blob_fallback_chop` in `tracknet_runner`, wired through WindowingPreset / IndexingConfig / the served path / `rally_seg_eval --blob-fallback`.

Why a factor gate, and why not just `window_hard_cap`: `hard_cap` chops **before** R4 and fires on *every* gap-less over-cap window ? it over-segments normal clips (mahadevpura_2 40?68). R5 runs **after** R4, and the magnitude gate (**default 4×**) leaves a normal long rally (~1?2× the cap) alone, force-cutting **only** the degenerate blob. That makes it surgical and do-no-harm.

MEASURED (n=11, on the cap6+R4 milestone, R2 `tolF1`):

clip	milestone	+ R5	windows (milestone ? +R5)
mahadevpura_1 (the blob)	**0.000**	**0.238**	2 (max 65.8 s) ? 32 (max 4.1 s), all `forced_cut`
mahadevpura_2 (also continuously-active)	0.122	0.180	42 ? 100
kushagra_singles	0.078	0.110	45 ? 71
Badminton_BXH_2	0.328	0.324	78 ? 92 (?0.004, noise)
(the other 7 clips)	?	*unchanged*	no-op (no group exceeds 4× the cap)

At factor=4, R5 touches **only** the 2 continuously-active clips + a noise-level BXH2 ?0.004 ? effectively **do-no-harm**. It is a **STRATUM tool**: the win is the blob rescue (0.000?0.238, removing the mean-swinging 0-match pathology), not a corpus-mean lift (agg `tolF1` 0.323?0.354, sign **3/3/5, n.s.** at factor 4; the factor sweep is monotonic ? 6× is strictly do-no-harm, 3× over-chops). **Default-OFF**; turn it on selectively for continuously-active footage. The R1 net over-seg guard confirms it: milestone?+R5 net cost 0.423?0.289, **PASS**, no merge_rate regression. [numbers: `scratch/r5_evidence.py`]

R1 ? milestone evidence (cap=6 + R4) + the net over-seg promotion guard *(2026-06-17)*
The R1 milestone is the smallest safe local-windowing flip: **W1 bounded windowing cap=6 + the R4 rally-END inactivity trim, no W2** (the per-clip-best config, settled by R3/R4). This entry (a) re-runs the full ablation ladder baseline?milestone under the **R2** `tolF1` on **n=11** with honest paired sign-tests, and (b) ships the **net over-seg promotion guard** that unblocks the milestone.

The full ladder (n=11, R2 `tolF1`, asymmetric ?_start=2/?_end=1.5):

22. assistant (2026-06-21T13:57:05.726Z)

I now have a comprehensive picture. Let me verify the §7 tracker freshness against current code/state ? specifically whether R2/R4 work is reflected in RALLY_QUALITY_RESEARCH.md §7 (it appears the §7 table still shows W2 net-crossings as the "next" item and n=6, while R1/R2/R4/R5/P7 have all shipped per QUALITY_ITERATIONS and the launch reports). Let me do a quick check on the gap-closure doc and confirm there are no other live tracker rows.

23. user (2026-06-21T13:57:06.506Z)

1 # Rally Detection ? Closing the Accuracy Gap (golden-eval-driven, per-video)
2
3 > **Status:** `DRAFT` ? implementation **owner-gated** · **Owner:** Avi · **Created:** 2026-06-18 · **Last updated:** 2026-06-18
4 > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)
5 > **Tracking anchors:** §7 progress tracker is the source of truth; pointer in memory `current-state`; add to `docs/README.md` + `docs/NEXT_STEPS.md` when work starts.
6 > **Relation to existing docs:** **peer-of** [`RALLY_QUALITY_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md) (that is the research + eval-framework doc; this is the *execution tracker* for the remaining accuracy gap). **Builds on** [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) (signal mechanics), [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md) (the R-log / empirical backbone), [`GOLDEN_VIDEOS.md`](data_pipeline/GOLDEN_VIDEOS.md) (the corpus). Does **not** duplicate them ? it points.
7 > **Honesty note:** each claim tagged `[verified]` (checked vs code/measured) / `[design]` (proposed). The headline numbers below are real measurements, but `n` is small ? read §6.
8
9 ---
10
11 ## 0. TL;DR
12 We are using the **growing golden corpus** (human rally labels) to find and close the rally-detection accuracy gap **per video, before launch** ? and the first ground-truth clip already reshaped the picture. The boundary work (R2/R3/R4) **holds** ? when local fires on a real rally, its edges are tight (|?| < 1 s). **The remaining gap is DETECTION, not boundaries:** (G1) **recall** ? both local and Gemini miss ~half the rallies on hard clips; (G2) **precision** ? local over-fires (false positives), especially on continuous drilling; (G3) **rally-END over-extension** ? when a player *quickly knocks the shuttle back after a point*, shuttle-velocity alone can't tell "rally" from "casual reset," so the window runs long / merges. n=2 sharpens the priority: **LOCAL = high recall / low precision, GEMINI = high precision / low recall** (opposite, complementary ? on the normal clip local's recall *beats* Gemini's). So **the #1 lever is precision** (cut local's false positives via `rally_gate` Gate-A + colocation; the player-state rally-END rule feeds this), with **signal fusion** (local's recall + Gemini-style precision) as the durable win. **Caveat:** two golden clips so far (n=15 corpus, GX010141+GX010137) ? directional, not conclusive.
13
14 ## 1. Problem & goal
15 The detector is good enough to ship a highlights product, but the golden eval shows clip-dependent accuracy that we want to **close consistently on every video, not just on average** ? catching the failure modes in the quality loop instead of after launch.
16
17 - **Concrete target:** on every corpus video, local is **at-par-or-better than Gemini vs the golden labels**, with **no per-video regression** on either eval set (the dual-eval-set discipline, [[eval-dual-set-regression]]).
18 - **"Good" =** the three gaps below closed enough that a milestone flip (default-ON) clears the over-seg guard AND improves per-video F1 broadly, recorded in a Launch Report ([[launch-report-convention]]).
19 - **Read first to get current:** [`RALLY_QUALITY_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md) §1?2 (the gap + diagnosis), [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) §4 (in-flight vs in-hand), [`HOW_RALLY_DETECTION_WORKS.md`](HOW_RALLY_DETECTION_WORKS.md) (the 2-layer model).
20
21 ## 2. Decisions locked
22 | # | Decision | Source / date | Implication |
23 |---|----------|---------------|-------------|
24 | D1 | The gap is **DETECTION (recall + precision + rally-END), not boundary precision** | golden GX010141, 2026-06-18 `[verified]` | Don't build a generic boundary head; spend on rally-state signals. |

D2	Measure **per-video AND aggregate; no regression on ANY video**	[[eval-dual-set-regression]] / [[decision-rigor-norm]]	A win must broadly help, not trade one clip for another.
D3	**Golden labels = ground truth; Gemini = reference, not truth**	n=15 corpus	Gemini under-detects too ? agreement ? accuracy. Report both, gate on golden.
D4	New signals land **default-OFF + flag-gated**, promoted only on measured lift	[`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md) / [[weak-signals-retained]]	Zero-regression rollout; retained even on a small-n null.
D5	**Rally-END needs a PLAYER-state signal** ? shuttle velocity can't separate a competitive rally from a casual post-point return	owner insight + R4 limitation, 2026-06-18 `[design]`	Add person kinematics / shuttle-in-hand as the rally-end discriminator.

3. Foundation ? the measured gaps (golden ground truth)
Ground-truth-scored Boxhill clips (more land as the owner labels them). Milestone windowing = cap6 + R4 ([`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md) R3/R4). Local & Gemini vs truth, n=2 so far: `[verified]`

clip (golden)	system	found	tIoU@0.5 P/R/F1	tolF1	hits	false+	missed	|?start|	|?end|
GX010141 (29, hard/drilling)	LOCAL	22	0.59/0.45/**0.51**	0.43	13	11	16	0.78	0.68
	GEMINI	16	0.94/0.52/**0.67**	0.53	15	4	14	0.31	0.46
GX010137 (34, normal match)	LOCAL	39	0.77/**0.88**/**0.82**	0.68	30	14	9	0.39	0.44
	GEMINI	27	1.00/0.79/**0.89**	0.89	27	0	7	0.50	0.52

- **Corpus aggregate (n=13, milestone):** tIoU F1@0.5 **0.474** [0.359, 0.579], F1@0.25 0.633; R2 tolF1 0.359. GX010137 (0.822) is the corpus's **best** clip; GX010141 (0.510) is above the mean. `[verified]`
- **Opposite failure modes ? the central finding (n=2).** **LOCAL = high recall, low precision; GEMINI = high precision, low recall.** On the normal clip GX010137, local's **recall 0.88 BEATS Gemini's 0.79** (local finds *more* real rallies) but local fires 14 false positives; Gemini fires **zero** false positives but misses 7 rallies. They are **complementary** (on GX010141 they overlap on only ~3 rallies yet each hits ~13?15 of the golden set). ? **fusion (local's recall + Gemini-style precision filtering) is the clear win.**
- **The gaps, re-prioritized by the data:**
  - **G2 ? precision is LOCAL's #1 gap** (over-fires on both clips: 11/22 and 14/39 false positives). This is the cheapest + highest-leverage fix (Gate-A / colocation). Filter local's FPs on GX010137 and it lands ~0.88+ ? at-par with Gemini, since local already out-recalls it.
  - **G1 ? recall** is clip-dependent: a local *strength* on normal play (? Gemini), but on hard continuous-drilling clips *both* miss ~half (GX010141 recall ~45?52%). The recall lever matters mainly for the hard clips.
  - **G3 ? rally-END over-extension** (quick-return-after-point): shuttle keeps moving ? window runs long / merges; R4 (shuttle-only) can't catch it. Contributes to G2's false positives/merges.
- **Local's gap to Gemini SHRINKS on normal play:** hard clip ??0.16 (0.51 vs 0.67) ? normal clip ??0.07 (0.82 vs 0.89), and local *out-recalls* Gemini there. The deficit is precision, not detection.
- **What is NOT the gap:** boundary precision. When local hits a rally its edges are tight (|?start|/|?end| ? ~0.8 s, on GX010137 even tighter than Gemini). The R2/R3/R4 windowing is doing its job. `[verified]`
- **Long-rally lens refinement (P7, n=15, `[verified]`):** filtering the eval to >5s rallies (the eye-catching highlights) shows local's single-court over-fire is **short-FP-dominated** (seg_ratio 1.43?1.04 ? duration-filterable), but **multicourt still over-fires 1.58× on LONG rallies** (adjacent-court games are genuine long rallies). ? the multicourt precision gap needs **court-localization (Gate-B / court mask, Lever C)**, NOT a duration filter; and the strict tolerance precision is gated by **over-merge contamination** of the long-pred set, not just short FPs. See [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md) "P7".
- ? **Data caveat:** GX010141 was labeled before ~3 rally-annotator-tool bugs were found + fixed (GX010137 is post-fix). GX010141's numbers may shift on a re-export ? re-verify before drawing hard conclusions from it.

4. Design / architecture ? the levers (REUSE MAP)

52 The signal mechanics live in [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md); this is which lever attacks which gap.
53
54 - **Lever A ? player-state rally-END rule** (targets **G3 + G2**) `[design]`. Generalize R4 from shuttle-only to **shuttle OR bilateral player motion**: end the rally at the last frame with competitive evidence from *either*, and **hard-terminate on a "shuttle picked up" event**. Two cheap detectors, no new model required:
55 - *Player kinematics* ? person velocity drops / players go stationary after a point (the casual returner taps it while the other walks). **Reuse:** the person cue producer (torchvision + ByteTrack) in `yolo_hybrid` (fusion P1, `[verified]` exists).
56 - *Shuttle-in-hand / out-of-play* ? `shuttle_player_colocation` (shuttle box ? person box = held) `[verified]` built in `fusion_features.py` (default-OFF, unmeasured on golden) from the fusion plan §4, **or** the no-new-model proxy: **sustained WASB `Visibility=0`** after the last fast stroke (held shuttle goes dark). Use it as a **termination anchor + reverse-timeline search** to the true end (last net-crossing / last fast-shuttle / last bilateral motion) ? the owner's idea, refined: the hold event is *late*, so anchor-then-backtrack, don't use its timestamp as the end.
57 - **Lever B ? recall** (targets **G1**) `[design]`. Both systems miss real rallies ? detector sensitivity: proposal-windowing / velocity-threshold tuning, and the **complementary-fusion** angle below. Measure recall floor per video; the highest-leverage labels are the *missed* rallies (active learning, RALLY_QUALITY_RESEARCH §4.3).
58 - **Lever C ? signal/ensemble fusion** (targets **G1**) `[design]`. Local & Gemini (and shuttle vs person cues) hit *different* rallies ? a union/ensemble lifts recall. Already-built scaffolding: `rally_gate.py` **Gate A** (net-crossings, structural rally-state, `[verified]` built, eval-only ? wire to prod = cheapest precision win for G2) and the FusionSegmenter feature path.
59 - **Pure-precision cleanup** (**G2**): `rally_gate` Gate A kills single-feed / held-shuttle *starts* structurally (?0?1 net crossings vs a real rally's several) ? the cheapest G2 win and independent of the person stack.
60
61 **Reuse map (what's new vs there):** `rally_gate.py::gate_windows` (Gate A, built) · `tracknet_runner` R4 end-trim (extend, don't replace) · `backend/eval/windowing.py` (preset knobs) · person producer in `yolo_hybrid` (built) · `shuttle_player_colocation` (built in `fusion_features.py`, default-OFF + unmeasured on golden) · `scratch/ingest_golden.py` + `scratch/boxhill17_score_golden.py` (the per-video golden scoring loop, `[verified]`).
62
63 ## 6. Honest limits ? what this does NOT do
64 - **n is small.** One golden clip scored so far (GX010141); the corpus is n=15 with several weak clips. Per-video conclusions firm up only as the owner labels more (GX010137 / GX020140 / a large-count clip next).
65 - **Nothing here is shipped or even started** ? implementation is owner-gated. A DONE row in §7 means *measured on golden*, not *flipped on by default*.
66 - **Player-state levers need GPU person-detection on the golden clips** (not yet extracted); until then Lever A/B are design, not data.
67 - The boundary tightness is from clips where local *hits* the rally; it says nothing about the rallies it misses (G1).
68
69 ## 7. Deliverables & progress tracker ? **source of truth**
70 Legend: ? Todo · ? In progress · ? Done · ? Gated. One small PR per row, off `origin/master`, no stacking.

24. assistant (2026-06-21T13:57:09.599Z)

Let me read the gap-closure §7 tracker (the execution tracker for the remaining gap) to capture the live next-step rows, since this is now the active execution doc peer to RALLY_QUALITY_RESEARCH.

25. user (2026-06-21T13:57:11.124Z)

69 ## 7. Deliverables & progress tracker ? **source of truth**
70 Legend: ? Todo · ? In progress · ? Done · ? Gated. One small PR per row, off `origin/master`, no stacking.
71
72 | ID | Deliverable | Depends on | Gated? | Status | PR |

|----|-------------|-----------|--------|--------|----|
P0	Golden scoring loop (`ingest_golden.py` + `boxhill17_score_golden.py`); GX010141 ingested (#12) + scored	?	No	?	this work
P1	Ingest the rest of the Boxhill golden batch (137/020140 + a large-count clip) ? per-video baseline table	owner labels	? owner	?	?
P2	Wire `rally_gate` **Gate A** (net-crossings) to the served path ? cheapest G2 win	?	? owner	?	?
P3	Extract person-velocity + `shuttle_player_colocation` on the golden clips (default-OFF features)	P1	? owner	?	?
P4	**Lever A**: player-state rally-END rule (R-next) ? measure ?F1 / ?merge-rate / ?|?end| vs R4	P3	? owner	?	?
P5	**Lever B/C**: recall + complementary fusion ? measure recall lift, no precision regression	P3	? owner	?	?
P6	Launch decision: milestone + best levers default-ON, with a Launch Report (both rulers + over-seg guard)	P4,P5	? owner	?	?
P7	**Long-rally (>5s) quality lens** (eval `--min-duration`/`--stratify`) + reel knob (`stitching.min_rally_duration`), both default-OFF. **Measured n=15:** single-court over-fire is short-FP-dominated (seg_ratio 1.43?1.04) but **multicourt over-fires 1.58× even on long rallies** (? court-localization, Lever C) and strict tolP doesn't rise (over-merge-contaminated). See [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md) "P7".	?	No (eval+cfg, default-OFF)	? measured	this PR

8. Open questions ? owner / external
1. **Which lever first** once labels land ? the no-model wins (Gate A wiring + shuttle-invisibility end-anchor), or go straight to person-velocity? *(decides: owner, after the per-video baseline)*
2. **Per-video target** ? "at-par-with-Gemini-vs-golden on every clip," or an absolute F1 floor? *(decides: owner)*
3. **Recall lever** ? detector-sensitivity tuning vs ensemble fusion as the primary path for G1. *(decides: data, post-P3)*

9. Definition of done
- [ ] Golden corpus covers the Boxhill set + enough variety that per-video reads are trustworthy (n meaningfully > 12).
- [ ] The **rally-END quick-return** failure mode (G3) is closed: measured ? merge-rate + ? |?end| on the clips that exhibit it, no regression elsewhere.
- [ ] Local is **at-par-or-better than Gemini vs golden on every corpus video** (no per-video F1 regression).
- [ ] The winning levers are flipped **default-ON** via a Launch Report (over-seg guard clean on both rulers).
- [ ] ? flip `Status ? DONE`, archive to `docs/archives/`.

10. References
Code anchors `[verified]`: `backend/pipeline/detectors/rally_gate.py` (Gate A) · `backend/pipeline/detectors/tracknet_runner.py` (R4 end-trim, `_trim_inactive_tail`) · `backend/eval/windowing.py` · `backend/eval/rally_seg_eval.py` + `tolerance_metrics.py` (R2) · `backend/pipeline/segmenters/yolo_hybrid.py` (person cue) · `scratch/ingest_golden.py`, `scratch/boxhill17_score_golden.py`.
Internal docs: [`RALLY_QUALITY_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md), [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) §4, [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md), [`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md), [`GOLDEN_VIDEOS.md`](data_pipeline/GOLDEN_VIDEOS.md), [`HOW_RALLY_DETECTION_WORKS.md`](HOW_RALLY_DETECTION_WORKS.md).

Changelog
- 2026-06-18 ? Created. Seeded with the GX010141 golden scoring (first ground-truth clip), the three named gaps (recall / precision / rally-END), and the player-state end-rule design from the owner's quick-return insight.
- 2026-06-18 ? Added GX010137 (golden #13, normal match play): local F1@0.5 **0.82** (recall **0.88 > Gemini 0.79**) vs Gemini 0.89. Confirms **opposite failure modes** (local high-recall/low-precision, Gemini the reverse) ? **precision is local's #1 lever**, fusion the durable win. Flagged the GX010141 pre-annotator-fix data caveat.

104   - 2026-06-18 ? **P7 long-rally (>5s) quality lens + reel knob (both default-OFF), measured n=15.** Eval `--min-duration`/`--stratify` (`rally_seg_eval.py`) + product `stitching.min_rally_duration`. Finding: single-court over-fire is short-FP-dominated (seg_ratio 1.43?1.04) but **multicourt over-fires 1.58× even on long rallies** (? needs court-localization, Lever C) and strict tolP is gated by over-merge contamination, not just FPs. Sharpened §3 + added §7 row P7.
105

26. assistant (2026-06-21T13:57:17.735Z)

I now have a complete and current picture of the rally-quality track across all the docs. I have everything needed to produce the structured output. Let me capture the staleness findings I noticed and assemble the answer.

Key findings on freshness: the RALLY_QUALITY_RESEARCH.md §7 tracker is partially stale ? it still shows n=6 and W2/multi-feature scorer as "next (GPU)", while QUALITY_ITERATIONS + R1_MILESTONE_LAUNCH_REPORT + GOLDEN_REGRESSION_FIXTURES show the corpus has grown to n=15, R1 milestone (W1 cap6 + R4) shipped default-ON (PR #204), R2 metric implemented, R5 blob-fallback + P7 long-rally lens shipped default-OFF, and the active execution tracker has moved to RALLY_DETECTION_GAP_CLOSURE.md §7.

27. user (2026-06-21T13:58:31.439Z)

Structured output provided successfully

28. assistant (2026-06-21T13:58:45.847Z)

I've completed the rally-quality track review and returned the structured findings. Summary of what I extracted:

Status: The rally-quality project's windowing wins are shipped. The R1 milestone (W1 cap=6 + R4 rally-END trim) is **flipped default-ON** (PR #204, baseline?milestone R2 tolF1 +0.222 at n=13, p=0.0005, over-seg guard PASS, independently reproduced). E1 baseline, W1/W1.5, W2 net-crossings first-step, R2 tolerance metric (augmenting tIoU), R5 blob-fallback, and the P7 long-rally lens are all landed (the latter three default-OFF). The full experiment/ablation harness exists. The active frontier moved to `RALLY_DETECTION_GAP_CLOSURE.md` §7, which re-prioritized the gap to **detection (precision #1, then recall, then rally-END)** with **Gate-A wiring** as the cheapest precision win.

Prioritized next steps (captured in the structured output): grow golden corpus to n?16 (highest leverage, owner-gated) ? wire Gate-A to prod ? extract person/audio features on golden clips ? W2 multi-feature learned scorer LOVO ? R2-vs-tIoU ablation to promote R2 as the gate + IAA ?-calibration ? Lever A (player-state rally-END rule) ? Lever B/C fusion ? M2 per-venue calibration ? finish Golden Regression Fixtures M4/P0. Tier-3 owned boundary-head is a corpus-gated stretch.

Key gates: most ML steps are **blocked on GPU-extracted golden features + labels (owner-side, not in this checkout)**; the gap-closure rows P1-P6 are **owner-approval-gated**; the ship-gate target (#172) is still an open owner decision.

Staleness flagged (not fixed): `RALLY_QUALITY_RESEARCH.md` §7 tracker still says n=6 and lists W2 as "next" while the corpus is n=15 and R1/R2/R5/P7 have shipped ? it should be reconciled or point to the now-live `RALLY_DETECTION_GAP_CLOSURE.md` §7. The `NEXT_STEPS.md` rally-quality detail body is dated 2026-06-15 (pre-milestone). Corpus `n` is cited inconsistently (6/13/15) across docs, and the headline gap-closure % is still uncomputed (Gemini not measured on the same golden set).

One process note: I could not `git pull --ff-only` ? it failed with "Cannot fast-forward to multiple branches," so I worked read-only against the local checkout; the docs may lag the remote.